

Pivoting away from isomorphism: Content, Sound, and Constituent Structure

We make the double claim that the domain of Content (‘lexical meaning’) is not a constituent, but a Span, and that the search for atomic Content is contingent on phonological representations. We argue for such a dependency based on properties of (so-called) Hebrew complex prepositions, such as *ʕal jed-ej* (*múziká*), ‘by (music)’ (lit ‘on hands of music’). The sequence P+N(+X) in Modern Hebrew (MH) as well as in Biblical Hebrew and interim periods comprises hundreds of attested tokens with rich idiosyncratic Content. However, the phonological, prosodic, Content, and syntactic properties of P+N(+X) combinations do not align, with N forming a constituent with its complement that excludes P (*jedej muzika*), but with Content and phonological properties associated with the non-constituent P+N (*ʕal jedej*) arguing for the syntactic structure [P [N DP/X]] in which Content can be associated with the non-constituent P-N, a Span. To affirm that conclusion, we discuss and dismiss on empirical grounds several alternative analyses, among which (i) the view that P+N are an instance of a possibly complex P; (ii) the possibility that N incorporates into P; (iii) the possibility that Content is associated not with the sequence P-N, but with N alone in the context of a designated P.

Keywords: Complex prepositions, Content, Modern Hebrew, Spans, Syntactic constituents, Pivots.

1 Introduction

To what extent does syntactic constituency define chunks of substantive word meaning (Content)? In an attempt to address this question, we focus on little-studied so-called complex prepositions in Modern Hebrew (MH) (henceforth PPiv for P+Pivot), and in particular on those taking a direct DP complement (see (1) for a sample):^{1,2}

- (1) a. *li=ʕn-ej* *ha=milħama*
to=FACE-MPL.CS DEF=war
‘before the war’
b. *bi=glal* *ha=ʕémeʃ*
in=TURD DEF=sun
‘because of the sun’
c. *bi=mkom* *ha=ħatuna*
in=PLACE DEF=wedding
‘instead of the wedding’
d. *mi=bħin-at* *ha=memʕala*
from=EXAMINATION-FSG.CS DEF=government
‘from the perspective of the government’

As we proceed to show, these complex expressions and many similar ones exhibit a mismatch between syntactic constituency on the one hand, and the domain of Content and at least some phonological realization on the other. The account for these mismatches would, in turn, lead us to endorse the following claim:

- (2) Content is matched neither with a syntactic constituent nor with a syntactic terminal, but with a *Span*, where a Span is a locally defined string of terminals (modified from Svenonius 2012, 2016).

PPivs abound in MH, taking DP, CP and PP complements, and occurring intransitively as well. We focus primarily on PPivs with DP complements because they are particularly structurally informative.

¹ Throughout, we use italics for Pivots (to be defined) and italicized small caps for their literal meaning.

To highlight pronunciation variations, transcriptions of MH words follow a minority Mizrahi pronunciation in marking the pharyngeals [ħ, ʕ]. The rhotic of that pronunciation (phonetic [r]) is transcribed as [r]. An acute accent indicates stress when it is not final. When there is a need to distinguish secondary and primary stress, the former is marked with a grave accent and the latter with an acute one, even when final.

² *Panim* (*pnej* in the plural Construct form), ‘face’, is pluralia tantum in Hebrew.

(3)	<u>Free form</u>	<u>Bound form (CS)</u>	<u>Gloss</u>	<u>Function</u>
a.	bhin-a	bhin-at	examination	FSG
b.	pan-im	pn-ej	face	MPL
c.	makom	mkom	place	MSG

(4) a. [DP₁ [N₁ mkòm [DP₂ ha=[N₂ ĥatuná]]]]
 place DEF=wedding
 ‘the place of the wedding’ (independently: makóm)

b. [DP₁ [N₁ pn-ěj [DP₂ ha=[N₂ jéled]]]]
 face-MPL.CS DEF=child
 ‘the face of the child’ (independently: paním)

c. [DP₁ [N₁ breχ-āt [DP ha=[N₂ kfár]]]]
 pool-FSG.CS DEF=village
 ‘the pool of the village’ (independently: breχá)

Nevertheless, and although the string <P, Pivot> is morpho-phonologically fixed and associated with its own Content, we will argue that it is neither a constituent nor a complex P head. Rather, and in line with standard assumptions about the structure of PPs with nominal complements as in (5)a, the structure of PPiv is as in (5)b, with the Pivot serving as a head of a Construct and thus, per force, failing to form a constituent with P. (5)b does differ from (5)a in one important respect: while in (5)a the Construct as a whole is a DP, we assume that in (5)b it is either a truncated nominal projection, N(or *n*), or a projection of PL.³ We argue for this claim and explore its consequences in section 4.

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$$(6) \quad \underbrace{[pP \underbrace{[N/n \ N \ \underbrace{[DP \ \dots N \dots]}_{Construct}}]}_{\langle P, PIVOT \rangle}]$$

In what follows we argue in some detail for the structure in (6). We conclude that the paradox can be resolved if one assumes that both Content and morpho-phonology for the PPivs are associated with the string $\langle P, Pivot \rangle$ because it forms a *Span*. Our proposal severs both Content and morpho-phonology from syntactic constituent structure.

The paper is structured as follows. In section 2, we provide general background, discussing the phonological properties of ‘simple’ prepositions, and the morpho-phonology and Content of both Pivots and PPivs. Section 3 provides primary categorial and morphological evidence that Pivots are nominal, while section 4 shows that while they are nominals, Pivots nonetheless are not DPs, and are rather a low nominal projection (NP or PL^{max}), thereby excluding individual reference and exhibiting, across the board, properties typically associated with pseudo noun incorporation (PNI) nominals. Section 5 highlights salient structural properties of Construct nominals, given their centrality to our argumentation, turning in section 6 to a distributional comparison between ‘simple’ Ps in MH and PPiv, with the aim of establishing that the two cannot be represented through an identical structure.

Sections 7 and 8 are devoted to our own analysis. Section 7 outlines our model of Content, focusing on its syntactic domain and its crucial interaction with phonological realization. Given our conclusion that PPivs are not constituents combined with the fact that they are nonetheless associated with atomic Content and with unique phonological realization, we argue in section 8 that PPivs are, rather, Spans, and that as a consequence, Spans must be allowed as appropriate domains not only for phonological realization (as has been argued already) but for Content matching as well. In proposing that Spans constitute Content domains together with their related phonological realizational properties, we depart from all previous Span accounts that we know of, which are focused exclusively on the realizational domains of functional strings.⁴

A challenge to our analysis could emerge, *prima facie*, from an alternative involving incorporation. Because the Pivot straddles two domains, one could propose that it merges first as a Construct head and then incorporates into P. Section 9 shows that given a well-constrained model of prosody, an incorporation analysis is impossible.

Section 10 closes the paper by highlighting, among other results, the conclusion that there is no uniform *intentional-conceptual* interface. Rather, syntax and formal semantics interface with an *intentional* component, while the *conceptual* component takes as its input annotated phonological representations (PR).

We have opted to relegate to the Appendix additional important structural differences between simple P and PPiv involving the choice of complementizers. While logically that discussion belongs in section 6 of this paper, its complexity would have taken us too far off our main narrative had we left it in that section.

2 Background

2.1 Simple prepositions

In the absence of substantial previous work and systematic dictionary listing, we have assembled our own corpus which includes (as of the completion of this paper) 230 PPivs. Of these, over a 100 take a DP complement.⁵

⁴ A potential exception is Preminger (2021), which is unfortunately too incomplete to allow a fuller evaluation.

⁵ Many PPivs take a CP complement as well as DP. Others take CP or PP only, and many are intransitive. A few examples are in (i). (✱: ‘cran’ form or obscure). For in-detail discussion of CP complements in both Ps and PPivs see appendix.

The 8 (simple) prepositions in MH are listed in (7). All are clitics. They are either bare consonants with epenthetic [e] or monosyllabic CV(C). The first five readily and productively participate in PPivs. In subsection 6.5 we argue that the list in (7) should be supplemented with a null preposition (P $\emptyset$):⁶

- (7) a. le/li ‘to’ c. ke/ki ‘as’ e. ʕal ‘on’ g. ʕim ‘with’
b. be/bi ‘in’ d. mi/me ‘from’ f. ʕad ‘until’ h. ʔel ‘toward’

2.2 The Content of Pivots

Following the P in PPiv is an item which we have dubbed *Pivot* (e.g., *glal* or *bxinat* in (1)). The Pivot, in turn, is followed by the DP complement in most cases under consideration here. The properties of Pivots will be of great interest to us in this study. (8) gives an overview of their Content possibilities. Examples are in (9) (recall: *Pivot* in italics, *LITERAL* gloss in italicized small caps).

- (8) a. Many Pivots are tokens of attested nouns found in isolation, but when combined with a fixed P in a PPiv, the resulting Content is distinct (9)a and non-compositional.
b. Some Pivots are otherwise unattested derived nominals, built of attested stems or roots with identical related or unrelated Content (9)b.
c. Some Pivots are altogether not attested outside PPiv contexts (they are, effectively, ‘cran’ forms, notated throughout as $\star$). Whatever Content they have is only available in conjunction with the relevant fixed P (9)c.
d. Finally, some Pivots have the very same Content that they would have in other nominal contexts (9)d. The emerging PPiv is nonetheless not fully compositional because the choice of P is arbitrary yet fixed, and crucial for the specific Content to emerge, (e.g., if *mi=* in (9)dii were to be replaced with *be=*, the ‘point of view’ Content would be lost)

(9)	<u>P=Piv</u>	<u>Complement</u>	<u>Gloss</u>
a. i.	le= <i>régel</i> to= <i>LEG, PILGRIMAGE</i>	(ha=)ʕiputs DEF=overhaul	‘on the occasion of (the) overhaul’
ii.	bi= <i>ʕvil</i> in= <i>PATH</i>	(ha=)ʕatula DEF=cat	‘for (the) cat’
iii.	mi= <i>pʔat</i> in= <i>EDGE, SIDEBURN</i>	(ha=)géʕem DEF=rain	‘because of (the) rain’
iv.	bi= <i>glal</i> in= <i>TURD</i>	gil-o age-POSS.3MSG	‘because of his age’
b. i.	be= <i>ʔemʕaʕ-ut</i> in= <i>CENTER-ABS.N</i>	patiʕ hammer	‘with a hammer’ cf. [_N <i>ʔemʕaʕ</i>] ‘center’ and [_N <i>ʔemʕaʕ-i</i>] ‘means’
ii.	le= <i>maʕbir</i> to= $\star$.N		‘abundantly’ cf. [_V <i>hiʕbir</i>] ‘increased’

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- (i) a. mi=*ʕum* ʕe be=*ʕod* ʕe (CP)
from=*ANY* COMP in=*MORE* COMP
‘because’ ‘while’
b. mi=*saviv* le= me=*ʕal* le= (PP)
from=*ROUND* to= from=*ABOVE* to=
‘around’ ‘above’
c. be=*séder* le=*vad* ($\emptyset$)
in=*ORDER* to= $\star$
‘alright’ ‘alone’

⁶ We follow inter alia Gesenius (1910) in taking the preposition *k=*, roughly ‘like’, to be homophonous with a complementizer, occurring, for instance in *k=ʔe=* ‘when’, *ka=ʔaʕer* ‘when’, as well as in apparent PPivs followed by passive participles, such as *ka=muʕan* ‘as is understood’, ‘of course’ or *ka=nirʔe* (‘as is seen’, ‘possibly’). See also fn.

iii.	be= <i>ʕiʕsum</i> =o _i in=★-OBJ.3MSG	ʕel (ha=)maʕber _i of (DEF=)crisis	‘in the middle of the crisis’ cf. [_N <i>ʕeʕsem</i>] ‘bone’, ‘essence’ ⁷
c. i.	be= <i>gin</i> in=★	gneva theft	‘because of a theft’
ii.	le= <i>ʕum-at</i> to=★-FSG.CS	sus horse	‘juxtaposed with a horse’
iii.	mi= <i>l=vad</i> from=to=★	ha=sus DEF=horse	‘except the horse’
d. i.	le= <i>ʕad</i> to= <i>SIDE</i>	ha=bájit DEF=house	‘on the side of the house’
ii.	mi= <i>ʕad</i> from= <i>SIDE</i>	ha=kala DEF=bride	‘of the bride’s side’; ‘from the bride’s point of view’
iii.	li= <i>jmin</i> to= <i>RIGHT</i>	ha=ʕofetet DEF=judge	‘to the right of the judge’ ‘in support of the judge’

Importantly, the same Pivot can be used with more than one P to give rise to different Content:

(10) a.	le= <i>vad</i> to=★ ‘alone’	bi= <i>l=vad</i> in=to=★ ‘only’	mi= <i>l=vad</i> from=to=★ ‘except’
b.	le= <i>fem</i> to=NAME ‘for the sake of’	be= <i>fem</i> in=NAME ‘on behalf of’	
c.	le= <i>jad</i> to=HAND ‘next to’	al= <i>jed-ej</i> on=HAND-MPL.CS ‘by’ (as in passive)	mi= <i>jad</i> from=HAND ‘immediately’

As we shall see, the syntactic and phonological properties of PPivs are constant across the board, and do not distinguish between those that are (semi) compositional as in (9)d and those that are not.

2.3 The morpho-phonology of Pivots

We noted that PPivs frequently occur in a morpho-phonological form which is distinct from that of non-pivotal P-N combinations. The differences primarily concern vowel raising for the prepositions *b-* and *l-*, and the spirantization of the first consonant of the Pivot (for *p*, *k*, *b*). These are obligatory for PPiv. Although they do otherwise exist in MH, they typically mark poetry or a higher register. Some examples are in (11), where ^H marks a higher register. Crucially, both standard and high register forms are available for compositional cases, but only the ‘high register’ equivalent is available in PPiv. Replacing it with the standard variety would lead directly to the loss of atomic Content:

⁷ Morphologically, *ʕiʕsum* has the form of a nominal derived from the 3rd verbal pattern (e.g., *kidem-kidum* ‘advance-advancement’). However, the putative source verb *ʕiʕsem* is not attested, and *ʕiʕsum* never occurs as singular. It is, however, attested in MH in the pluralia tantum *ʕiʕsumim* ‘sanctions’, having Content which is entirely unrelated to that of (9)biii. Although plausibly related to the Content of *ʕeʕsem* ‘bone’, ‘essence’, that (putative) relatedness is clearly not derivational.

Note that *be=ʕiʕsum-o* ‘in the middle of’ *must* be associated with a clitic pronoun which is coreferential with its direct complement, giving rise to clitic doubling. Pronominal clitics are obligatory on several other Pivots, with either anaphoricity resulting as in (i), or as a fixed 3SG ‘pleonastic’ (ii). Our account has little to say about these beyond observing that syntactically, the clitics are direct complements, therefore clustering with our other DP-taking PPivs.

i. be= <i>gap</i> =i in=LIMB=1SG ‘alone’	be= <i>hub</i> =am be=★=3PL ‘in their heart’	be= <i>ʕetj</i> =a/o in=★=3FSG/3MSG ‘because of him/her’ (=it’s her fault)‘	ii. bi= <i>zman</i> =o/*a/*am in=TIME=3MSG/3FSG/3MPL ‘once (past reading)’
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(11)	<u>Free form</u>	<u>Bound CS form (compositional)</u>	<u>PPiv bound Pivot</u>
a.	le=pe to=mouth 'to a mouth'	le=pi/pe le= ^H fi ^H to=mouth.MSG.CS DEF=horse 'to the horse's mouth'	*le=pi/pe ha=sus le=fi to=MOUTH.MSG.CS DEF=horse 'according to the horse'
b.	le=panim to=FACE 'to a face'	le=pn-ej li= ^H fn ^H -ej to=face-MPL.CS dog 'to a dog's face'	*le=pn-ej kélev li=fn-ej to=FACE-MPL.CS dog 'before a dog'
c.	le=kavod to=honor 'to honor'	le=kavod / le=kvod ha=mifpaḥa li= ^H χvod ^H to=honor DEF=family 'for the family's honor'	*le=k(a)vod ha=mexira li=χvod to=HONOR DEF=sale 'in anticipation of the sale'

3 Pivots are nominal⁸

3.1 Bound Construct form

As noted (cf. (3)) and as is well-established, in N₁+N₂ Constructs in MH the syntactic head N₁ occurs in some contexts in a bound morpho-phonological form. Specifically, feminine singular (FSG) occurs as *-a* in free forms, but as *-at* in bound forms. Masculine plural (MPL) occurs as *-im* in free forms, but as *-ej* in bound forms. And finally, syncope impacts some (but not all) /a/ vowels in unstressed open syllables in N₁ (see section 9 for a detailed discussion). A few additional bound Construct examples are in (12)-(14) alongside their free forms:

(12)	<u>Construct-MPL</u>	<u>Free form-MPL</u>	<u>Gloss</u>
a.	sifr-ej ha=sifrija book-MPL.CS DEF=library	ha=sfar-im (fel ha=sifrija) DEF=book-MPL (of the library)	'the books of the library'
b.	ḥatul-ej ha=kafe cat-MPL.CS the=café	ha=ḥatul-im (fel ha=kafe) DEF=cat-MPL (of the café)	'the cats of the café'
c.	jald-ej ṣaza child-MPL.CS Gaza	ha=jelad-im fel ṣaza DEF=child-MPL of Gaza	'the children of Gaza'
(13)	<u>Construct-FSG</u>	<u>Free form-FSG</u>	<u>Gloss</u>
a.	brex-at ha=ṣir pool-FSG.CS DEF=city	ha=brex-a (fel ha=ṣir) DEF=pool-FSG (of DEF=city)	'the city's pool'
b.	kalb-at ha=mifpaḥa dog-FSG.CS the family	ha=kalb-a (fel ha=mifpaḥa) DEF=dog-FSG (of DEF=family)	'the family's dog'
c.	haftsats-at ṣaza bombing-FSG.CS Gaza	ha=haftsats-a (fel ṣaza) DEF=bombing-FSG (of Gaza)	'the bombing of Gaza'
(14)	<u>Syncope, bound form</u>	<u>Free form</u>	<u>Gloss</u>
a.	pkid ha=mas clerk.CS DEF=tax	ha=pakid (fel ha=mas) DEF=clerk (of DEF=tax)	'the tax clerk'

⁸ Within approaches that considers PPivs complex prepositions (i.e. of the category P in their totality), at least some of the nominal properties of the Pivots have led to the conclusion that Hebrew prepositions are nominals (Rosen 1966), or derived from nominals (Gesenius 1910; Ben-Asher 1974; Schleizer 1990, 1998 among others), alongside the claim that the category P allows productive word formation operation in Hebrew (in particular, Schleizer 1998). While we do, of course, subscribe to the view that the Pivot is nominal, we differ from such 'complex P' accounts in assuming that the Pivot is *not* part of a complex P, but rather, it heads its own nominal phrase.

At least one problem with the claim that prepositions in Hebrew are nominal (or derived from nominals) is the fact that undisputed 'simple' prepositions (that is, the list in (7)) never exhibit any such nominal properties when not accompanied by a Pivot. For a detailed contrastive comparison between 'simple' P and PPivs, see section 6 as well as the Appendix.

- | | | | |
|----|--|---|--------------------------|
| b. | m(e)kom ha=sifrija
place.CS DEF=library | ha=makom fel ha=sifrija
DEF=place (of DEF=library) | ‘the library’s location’ |
| c. | sviv ha=bájit
round.CS DEF=house | mi=saviv (l=a=bájit)
from=round to=DEF=house | ‘around the house’ |

When Pivots enter a Construct relation with (the syntactic head) of the DP complement, they acquire the bound form associated with the N head of the Construct, thereby signaling their nominal status:

- | | | | | | |
|------|---------|--|---------------------------|---------------------|------------------------------|
| (15) | | <u>P=Piv</u> | <u>Complement</u> | <u>Free form</u> | <u>Gloss</u> |
| a. | FSG | i. me= <i>ham-at</i>
in= <i>SUN-FSG.CS</i> | ha=safek
DEF=doubt | ham-a | ‘due to the doubt’ |
| | | ii. le= <i>tov-at</i>
to= <i>FAVOR-FSG.CS</i> | ha=sus
DEF=horse | tov-a | ‘for the good of the horse’ |
| b. | MPL | i. mi= <i>pn-ej</i>
from= <i>FACE-MPL.CS</i> | sakana
danger | pan-im | ‘from/because of danger’ |
| | | ii. <i>ʕal=jed-ej</i>
on= <i>HAND-MPL.CS</i> | namer
tiger | jad-áj-im | ‘by a tiger’ |
| | | iii. le= <i>gab-ej</i>
to= <i>BACK-MPL.CS</i> | ʕlomo
Shlomo | gab-ot ⁹ | ‘as for Shlomo’ |
| c. | Syncope | i. bi= <i>mkom</i>
to= <i>PLACE.CS</i> | ʕlomo
Shlomo | makom | ‘instead of Shlomo’ |
| | | ii. li= <i>jmin</i>
to= <i>RIGHT.CS</i> | ha=plitim
DEF=refugees | jamin | ‘in support of the refugees’ |

3.2 Pivots: gender and number

Like nominals, Pivots observe a gender distinction, and to the extent that they are tokens of otherwise attested Ns, share with those Ns the F/M feature. More significantly, Pivots may be both singular and plural. The selection for a plural or singular Pivot appears to be arbitrary, insofar as little meaning can be attributed to it. However, the choice is fixed relative to Content: forms occurring with a singular Pivot would lose their PPiv atomic Content if the Pivot is pluralized, and plural Pivots, similarly, must remain plural for the PPiv Content to be preserved. Some minimal pairs are in (16). Additional plural forms (without an attested singular PPivs) are in (17) (including some that are otherwise intransitive or take a PP complement).¹⁰

- | | | |
|------|--|---|
| (16) | <u>SG Pivot</u> | <u>PL Pivot</u> |
| a. | <i>ʕal=jad</i>
on= <i>HAND</i>
‘next to’ | <i>ʕal jed-ej</i>
on <i>HAND-MPL.CS</i>
‘by’ (passive) |
| b. | (be=) <i>ʕékev</i>
(in=) <i>HEEL</i>
‘as a result of’ | be= <i>ʕikv-ot</i>
in= <i>HEEL-FPL</i>
after, in the footsteps of |
| c. | (be=) <i>sviv-at</i>
(in=) <i>AREA-FSG.CS</i>
‘in the area of’ | be= <i>sviv-ot</i>
in= <i>AREA-FPL</i>
‘around’ (approximate) |
| d. | le= <i>régel</i>
to= <i>FOOT</i>
‘on the occasion of’ | le= <i>ragl-ej</i>
to= <i>FOOT-MPL.CS</i>
‘at the foothills of’ |

⁹ In the PPivs *ʕal=gab-ej* ‘on top of’ and *le=gab-ej* ‘concerning’ (lit. ‘on backs’ ‘to backs’) the masculine bound plural form *gab-ej* deviates from the standard form in compositional contexts, where the plural marker is *-ot*. Hence, *ʕal gab-ot*; *le gab-ot* ‘on backs’ ‘to backs’) when compositional. The MPL form does occur in older layers of Hebrew but is absent from contemporary MH. This ‘archaic’ morphological form is characteristic of the non-standard morpho-phonology frequently associated with PPivs already discussed in section 2.3.

¹⁰ None of the Pivots in our corpus are *Content M/F pairs*, making the preservation of Gender impossible to test.

- | | | | |
|----|--|--|---|
| e. | $\eta af = \varsigma al = gav$
NEG=on=BACK
'despite' | $\varsigma al = gab-ej$
on=BACK-MPL.CS
'on top of' | $le = gab-ej$
to=BACK-MPL.CS
'concerning' |
| f. | $le = \varsigma et$
to=TIME
'at the time of' | $le = \varsigma it-im$ (intrans)
to=TIME-MPL
'sometimes' | |
- (17) a. $hod-ot^{11}$ $le =$ (PP complement)
 $\star-FPL$ to=
'thanks to'
- b. $(\varsigma al) = \eta od-ot$
on= $\star-FPL$
'about'
- c. $li = frak-im$ (intrans)
to=SEGMENT-FPL
'intermittently'

3.3 Pivots and N categorial markers

We noted already (cf. (8)b that the Pivot could occur as a derived nominal, at times one which is not otherwise attested (e.g., $\eta em\tau a\varsigma ut$, the otherwise unattested abstract nominal derived from the noun $\eta em\tau a\varsigma$ 'center'). A few other categorially marked Ns are in (18):

- (18) a. $li = fkud-at$ $\eta lomo$
to=ORDER-FSG.CS Shlomo
'to the order of Shlomo (as in check)'
source verb: $pakad$ 'order'
- b. $be = \varsigma ezr-at$ $\varsigma amit$
in=HELP-FSG.CS colleague
'with the help of a colleague'
source verb: $\varsigma azar$ 'help'
- c. $be = mahala\chi$ $ha = jom$
in=GEAR/MOVEN DEF=day
'during the day'
- d. $le = \tau smit-ut$
to=SERF-ABS.N
'forever (lit. to=SERFHOOD)'

Additional evidence for the nominality of Pivots comes from their occurrence in exclusively nominal templates, where by 'template' we refer to patterns that are associated with a fixed major category as well as a rigid prosodic structure and a specific vocalization. The CCuCa form, attested in (some) derived nominals and exemplified in (18)a, as well as the mVCCaC form attested in (18)c are already such templates. Undersived ones are the Segolate templates CéCeC/CáCaC/CóCeC with their characteristic penultimate stress, which are always nominal. A short list of Segolate Pivots is in (19):

- | | | | |
|---------|---|----|---|
| (19) a. | $be = géder$
in=FENCE
'can be viewed as' | f. | $le = régel$
to=LEG
'on the occasion of' |
| b. | $be = kérev$
in= $\star$
'among' | g. | $be = mé\varsigma e\chi$
in=DURATION
'during' |
| c. | $mi = tá\varsigma am$
from-TASTE/STRESS
'as delegated by' | h. | $be = \varsigma ére\chi$
in=VALUE
'approximately' |
| d. | $ke = néged$
as=OPPOSITE
'as opposed to' | i. | $be = séder$
in=ORDER
'all right' |

¹¹ Related forms that do occur are *toda-todot* 'thanks.SG-thanks.PL' as well as *hodaja-hodajot* 'thanking.SG-thanking.PL'.

e. le=máʃan
to=ADDRESS
'for'

j. be=ʃétsem
in=BONE
'actually'

4 But Pivots are NOT DPs

4.1. Nominal Constructs without DP

A cornerstone of our proposal is the claim that the structure of PPiv is (in essence) identical to the structure of compositional PPs, schematically as in (20), with DP1 the complement of P, and DP2 the complement of N1, and with N1 and N2 forming a Construct. A very similar structure, we propose, is to be associated with PPiv, with the single difference that the complement of P is not a DP, but a lower nominal projection containing NP/nP and whatever syntactic node controls plural marking (call it PL^{max}) as in (21):

(20) [P P [DP1 N1 . [DP2 N2]]]
ʃal kalb-ej ha=ʃxuna
about dog-MPL.CS DEF=neighborhood
'about the neighbourhood's dogs'

(21) [P P [N/PL1 N- Pivot1 [DP2 N2]]]
ʃal jed-ej ha=ʃxuna
on HAND-MPL.CS DEF=neighborhood
⏟
'by' ('the decision **by** the neighborhood')

As a projection of N/PL, the properties of the Pivot are expected to mirror those of other so-called 'pseudo-incorporated' nominal (PNI) structures.¹² In the remainder of this subsection, we review the evidence for this 'truncated' Pivot+N structure, comparing it to that of full DP Constructs.

A brief preliminary comment is in order. Alongside DP-DP Constructs, MH also has DP-N/PL^{max} Constructs, labelled M-Constructs in Borer (2012), where the *complement* exhibits PNI diagnostics and is interpreted, as is to be expected, as a kind or as a property, and not as an individual (see next subsections for ample exemplification). As we proceed to show, the properties of these N/PL^{max} complements tally with those of Pivots. We note that Constructs headed by a nominal property, our Pivots, have not been previously proposed to the best of our knowledge.¹³ In subsections 4.2-4.6 we proceed to compare NP-DP Constructs (Pivot-DP), with both DP-DP Constructs (R-Constructs) and with DP-NP Constructs (M-Constructs).

4.2 The distribution of the definite article

In all variants of the Construct in MH only one instance of the definite article, *ha=*, is ever possible regardless of the complexity of the Construct. Standardly, it occurs on the *complement* N. With the definite article in that position, the DP-DP R-Construct in its entirety is definite, as is the complement (22)a. In DP-NP M-Constructs, on the other hand, the entire Construct is definite, but the complement may be non-referential (22)b (and hence definiteness for it is vacuous). Finally, in NP-DP Pivot Constructs the complement is definite, but the head is non-referential: (22)c:

¹² Term originally from Massam (2001), on the behavior of nominal complements within fronted VPs in Niuean. See also Massam (2017). On non-referential instances of nominals in MH see in particular Danon (2001). On PNI diagnostics in nominal complements in the Construct, see Borer (2012).

¹³ [Property]-DP Constructs *are* attested, providing the property, as well as the Construct in its entirety, is A, not N:

i. khol ʃalim ʃvur lev ʃav-at parva
blue leaves broken heart thick-FSG.CS fur
'of red leaves' 'of broken heart' 'of thick fur (f.)'

(22) DEF N2, distribution across Construct types

- a. R-Construct: *be=fem ha=jalda ha=zot*
in=name DEF=girl DEF=this ‘name’ DEF ; ‘girl’ DEF
‘(there is a spelling error) in the name of this girl’
- b. M-Construct: *be=fem ha=jalda ha=nadoj ha=ze*
in=name DEF=girl DEF=corny DEF=this ‘name’ DEF ; ‘girl’ NON-REF
‘(what do you see) in that corny girl’s name’
- c. Piv-Construct: *be=fem ha=jalda*
in=NAME DEF=girl ‘name’ NON-REF; ‘girl’ DEF
(they are acting) ‘on behalf of the girl’

Alongside the placement of the definite article *ha=* on the complement N in Constructs, contemporary MH allows its placement on N1. In these cases, N2 is, across the board, non-referential. As a consequence, while the placement of *ha=* on N1 is possible for M-Construct, it is impossible for R-Construct. Finally, Pivots, instances of N1, are rarely associated with DEF and never when they take a DP complement.¹⁴

(23) DEF N1 distribution across Construct types

- a. R-Construct: **b=a=fem jalda ha=zot*
in=DEF=name girl DEF=this
(there is a spelling error) in the name of this girl’
- b. M-Construct: *b=a=fem jalda ha=nadoj ha=ze*
in=DEF=name girl DEF=corny DEF=this ‘name’ DEF; ‘girl’ NON-REF
(what do you see) ‘in that corny girl’s name’
- c. Piv. Construct: **b=a=fem (ha)=jalda*
in=DEF=NAME DEF=girl
(they are acting) ‘on behalf of the girl’

We will assume, alongside too much research to cite, that the availability of a definite reading as well as the availability of individual reference necessitate D. We will further assume that in the absence of D (and Q/Num) what emerges (in compositional contexts) is a property/kind reading. The property reading for the complement of Constructs with the structure DP-NP follows directly. In the

¹⁴ There are altogether 4 cases of PPiv in our corpus which seem to involve a definite Pivot. All cases are strictly intransitive, with the definiteness semantically vacuous or weak:

- i. *l=a=xen b=a=zman l=a=rov k=a=fura*
to=DEF=SO in=DEF=TIME to=DEF=MAJORITY as=DEF=LINE
‘therefore’ ‘on time’ ‘mostly’ ‘appropriately’

While the transitive Pivot in (ii) may appear definite, in actuality the vowel of *be-* has undergone lowering preceding the pharyngeal. Structurally, if *ʕavur* were definite, it would give rise to an impossible DEF-DEF Construct. Similarly, we assume that lowering, rather than the definite article is responsible for the form *la* in (iii). Similar effects are note that similar effects are attested with the *l-* marker in infinitives (iv):

- ii. *ba=ʕavur ha=mexonit*
n=★ def=car
‘in exchange of the car’
- iii. *la=ʕalutin ka=ʕet la=ʕad*
to=★ in=time to=eternity
‘therefore’ ‘now’ ‘forever’
- iv. a. *li=tfor li-kro li-ʕmor*
to-sew to-read to-guard (infinitival forms, template I)
- b. *la=ʕavod la-xlom*
to-work to-dream

Note that in (iii), *ʕad* ‘eternity’, is a Pivot, although it also occurs as a simple P. See fn. 33 for some discussion of this issue.

In section 8.1 we briefly compare MH Pivots with English Pivots in so-called 3-word prepositions where, with a complement, the definite article is more common, although still not the norm (*at the behest of, for the sake of*, and others) and where the constraints on the definite article attested in the Contrast are, of course, irrelevant.

absence of compositional reading for (P) Pivot-DP configurations, we can safely exclude individual reference and hence D and Q/Num. Beyond that, assigning any semantics to the Pivot away from the accompanying P is impossible.¹⁵

4.3 Demonstratives

In R-Constructs, both nouns can be modified by a demonstrative (24)a-c, where the demonstrative may agree in gender with either DP, and the result is grammatical either way. As (24)b shows, in M-Constructs, the DP, but not the NP, can be modified by a demonstrative. Finally, for Pivot-Constructs, as in (24)c, the complement can correspond to a demonstrative, but not so the Pivot:

- (24) a. R-Construct
 i. be=fem ha=jalda ha=ze
 in=name^M DEF=girl^F DEF=DEM^M
 ‘in this name of the girl’
 ii. be=fem ha=jalda ha=zot
 in=name^M DEF=girl^F DEF=DEM^F
 ‘in the name of this girl’
 b. M-Construct (N1 DEF)
 i. b=a=fem jalda ha=ze
 in=DEF=name^M girl^F DEF-DEM^M
 ‘in this [girl’s name]’
 ii. *b=a=fem jalda ha=zot
 in=DEF=name^M girl^F DEF=DEM^F
 ‘in [this girl’s] name’
 c. Pivot-Constructs
 i. be=fem ha=jalda ha=zot
 in=name^M DEF=girl^F DEF=DEM^F
 ‘on behalf of this girl’
 ii. *be=fem ha=jalda ha=ze
 in=DEF=name^M DEF=girl^F DEF=DEM^M
 ‘on this behalf of the girl’

For the same reason that Pivots and complements in M-Constructs cannot host a definite article, they cannot be associated with a demonstrative as in (24)b,c. Both facts follow from the absence of D (and Q/Num).

4.4 Pronominal reference

In the absence of D, we should not be surprised to find that a pronoun may not refer back either to the complement in M-Constructs (25)b, or, indeed, to the Pivot (25)c. No such restriction applies to either Construct member in the R-Construct (25)a.

- (25) a. R-Construct
 be=fem ha=jalda je=?ot-o/-a hu baħar
 in=name^M DEF=girl^F COMP=PRON.ACC-M/F he chose
 ‘in the name of the girl that he chose (it^M/her^F)’

¹⁵ A comparable conclusion applies to so-called 3-word prepositions in English, such as *by virtue of* or *in dint of*. While both *virtue* and *dint* can be shown distributionally and morphologically to be nominals, and while it is safe to say that they are *not* individual references, it appears clear that the preceding P is crucial for whatever Content or meaning they acquire. Assuming that either *virtue* or *dint* in isolation are properties or kinds, appears at best circular, and at worst nonsensical. For a more detailed comparison between PPivs and 3-word English prepositions see section 8.1. For some comment on how to model specifically the Content dependence of P-Pivot or English 3-word prepositions, see fn.38 as well as section 8.2.

- b. M-Construct
 b=a=fem jalda fe=?ot-o/*-a hu baħar
 in=DEF=name^M girl^F COMP=PRON.ACC-M/F he chose
 ‘in the name of the girl that he chose (it)’
- c. Pivot-Construct
 be=fem ha=jalda fe=?ot-o^M/-a^F hu baħar
 in=NAME^M DEF=girl^F COMP=PRON.ACC-M/F he chose
 ‘on behalf of the girl that he chose (her)’

4.5 Adjectival modification

Like demonstratives and pronominal reference, adjectives may be associated with either DP in R-Constructs. In M-Constructs, modification of the NP is restricted (as expected) to Kind modification. Pivots, finally, cannot be modified by an adjective altogether:

- (26) a. R-Construct
 be=fem ha=?ela ha=mít-it /ha=míti
 in=name^M DEF=goddess^F DEF=mythical-F/ DEF=mythical.M
 ‘in the name of the mythical goddess’ / ‘in the mythical name of the goddess’
- b. M-Construct
 b=a=fem ?ela ha=míti (ha=ze)
 in=DEF=name^M goddess^F DEF=mythical.M (DEF=DEM^M)
 ‘in that mythical name of a goddess’
- c. Pivot-Construct
 be=fem ha=?ela ha=mít-it/*ha=míti
 in=NAME^M DEF=goddess^F DEF=mythical-F/*DEF-mythical.M
 ‘on behalf of the mythical goddess’

4.6 Plural marking

In R-Constructs plural marking occurs freely on the Construct head. In turn, as noted already (see subsection 3.2), while both plural and singular Pivots are attested, singular Pivots cannot be freely pluralized without loss of atomic Content, and vice versa – plural Pivots cannot occur as singulars without loss of Content:¹⁶

- (27) a. R-Constructs
 (hajta tsaléket) ʕal=jad/jed-ej ha=jalda/ha-jelad-ot (ha=zot/ha-?ele)
 (was scar) on=hand/hand-MPL DEF=girl/DEF=girl-PL (DEF=DEM.FSG/DEF=DEM.FPL)
 ‘(There was a scar) on the hand/s of the girl/s’
- b. Pivot Construct
 i. ʕal=jed-ej ha=jalda/ha=jelad-ot
 on=HAND-MPL.CS DEF=girl/DEF=girl-FPL
 ‘by the girl(s)’ (primarily in passive context)

¹⁶ M-Constructs are set aside in this subsection as they constitute a more nuanced case in ways that are orthogonal to our main narrative. Specifically, like Pivots, count complements of M-Constructs exhibit a fixed preference for singular or plural, but unlike Pivots, the alternatives, while dispreferred, are grammatical and interpretable (i-ii). A compounding factor is a number harmony of sorts which results in the complement – lacking individual reference, recall - agreeing with the head in SG/PL marking (iii) (see Faust 2021 for some discussion).

- | | |
|--|---|
| i. a. mits tapuz-im (#mits tapuz)
juice orange-PL (juice orange-SG) | b. mits limon (#mits limon-im)
juice lemon.SG (juice lemon-PL) |
| ii. a. fëmen zajit (#zejt-im)
oil olive.SG (olive-PL) | b. fëmen ħamanij-ot (#ħamanij-a)
oil sunflower-PL (#sunflower-SG) |
| iii. a. mitle kir (#mitle kiro)
hanger.SG wall.SG (hanger-SG wall-PL) | b. mitl-ej kir-ot (#mitl-ej kir)
hanger-PL wall-PL (hanger-PL wall-SG) |

- ii. $\zeta_{al}=jad$ $ha=jalda/ha=jeladot$
 on=*HAND* DEF=*girl/DEF=the girls*
 ‘next to the girl(s)’

The totality of the diagnostics in this section confirm the status of the Pivot as PNI and thus not a referential DP but rather a projection of N/PL. We will adopt this conclusion in the rest of this paper.

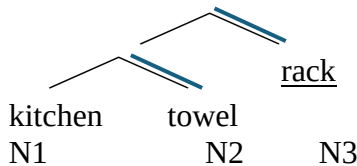
5. The architecture of Construct nominals

The reasoning under development in this paper cannot be complete without a better understanding of some crucial properties of the Construct.¹⁷ As we already pointed out (subsection 3.1), the head of the Construct must c-command its complement and be adjacent to it. Even more strictly, we suggest, the relationship between the head and its complement is that of *H-Government* (see section 8 for additional discussion):

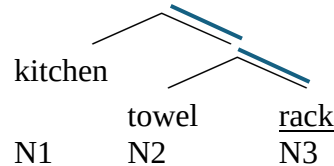
- (28) *H-Government*¹⁸
 $\alpha^{\min}$ *H-Governs* $\beta^{\min}$ iff $\alpha^{\min}$ is a sister of $\beta^{\max}$ and there is no χ s.t. $\alpha^{\min}$ c-commands χ and χ c-commands $\beta^{\min}$ and χ is phonologically overt.

Direct evidence that adjacency does *not* suffice to give rise to the Construct and that the relationship between the Construct head and its complement is delimited by H-Government is available from N1-N2-N3 strings. Such strings are ambiguous, contingent on whether the bracketing is $[[N1\ N2]\ N3]$ or $[N1\ [N2\ N3]]$, as illustrated by the classical English examples (29)a-b (syntactic head underlined, head projection nodes doubled). What is of some interest specifically for (29)b is the fact that the embedded compound, *towel rack* and the entire DP, *kitchen towel rack* are both headed by the same terminal at the right edge, and that this head (N3) does not c-command N1 which therefore cannot be a complement and must be an adjunct or a specifier. This structural configuration emerges as a direct result of the fact that in English compounds the head absolutely must be at the right edge (double bars mark headedness):

- (29) a. $[[kitchen\ towel]\ rack]$



- b. $[kitchen\ [towel\ rack]]$



N1-N2-N3 strings occur in MH as well (typically involving DP-NP Constructs), albeit with a *left-edge* syntactic head. Apart from the mirror image, the relationship between the headedness and the branching possibilities are identical in MH to those of English, as is to be expected. What is, however, of note, is that in structure (30)b, the correlate of (29)b, Construct formation between N2 and N3 is impossible. As predicted, cases with the structure in (30)a exhibit bound form for both N1 and N2. Cases with the structure in (30)b, however, sport bound morphology on N1, but not on N2 (see Borer

¹⁷ It is not our intention to offer here a complete analysis of the Construct, but rather to highlight some aspects of its structure which are crucial to our claim that the Pivot is a Construct head, and that PPivs are *not* constituents. For detailed analyses of the Construct the reader is referred to the massive literature on the topic, too extensive to list here.

¹⁸ Given the sisterhood condition built into H-Government, an intervening χ could not be a head, but only a specifier or an adjunct. We set aside for reasons of space the question of whether such specifiers or adjuncts are ever architecturally available, only to note that if not so, the definition of H-government could be considerably simplified.

For direct empirical support for preferring *sisterhood* in the definition of H-government over c-command, the reader is referred to section 9, and specifically to our conclusion that a weaker condition would fail to properly derive the prosodic properties of the Construct.

2012 for discussion.)^{19, 20}

- (30) a. [N1 [N2 N3]]

 sleeves shirt wool
 ‘sleeves of a woolen shirt’
- b. [[N1 N2] N3]

 sleeves shirt wool
 ‘shirt sleeves of wool’
- (31) a. [siml-at [pirh-ej méfi]]
 dress-FSG.CS flower-MPL.CS silk
 ‘dress of silk flowers’
- b. [farvul-ej [hults-at tsémer]]
 sleeve-MPL.CS shirt-FSG.CS wool
 ‘sleeves of woolen shirt’
- (32) a. [[siml-at [prah-im]] méfi]
 dress-FSG.CS flower-MPL silk
 ‘flowery dress of silk’
- b. [[farvul-ej [hults-a]] tsémer]
 sleeve-MPL.CS shirt-FSG wool
 ‘woolen shirt sleeves’

In (30)b, N2 cannot construct with N3 despite linear adjacency, despite being contained in the same DP, and despite the availability of M-command. This result follows directly, of course, if Construct formation requires sisterhood as well as adjacency (i.e. *H-Government*).

For completeness, note that where Construct formation is licit, it is obligatory, and that its presence forces a particular constituency, as follows from H-Government, and a particular interpretation. Thus in (33)a, because *miḡs-ej* ‘lids’ is bound, *plástik* ‘plastic’ must be its complement, yielding the interpretation of ‘plastic lids’. By the same token, because *plástik* ‘plastic’ is the complement of ‘lids’, the latter must be bound. Ipso facto, because in (33)b *miḡs-im* ‘lids’ is free, *plástik* cannot be its complement, and by the same measure, because ‘plastic’ is not a complement, ‘lids’ cannot be bound. The interpretational result now follows from scope, with N3 in (33)a interpreted, indeed, as a complement of N2, but in (33)b as a modifier of the entire phrase – ‘plastic [lid box]’, where it is the box that is plastic, but the lids could be of any material at all.

- (33) a. [kufs-at [miḡs-*im/ej plástik]]
 box-FSG.CS lid-MPL/MPL.CS plastic
 ‘a box of plastic lids’
 *‘a plastic box of lids’
- b. [[kufs-at [miḡs-im/*ej]] plástik]
 box-FSG.CS lid-MPL/MPL.CS plastic
 ‘a plastic box of lids’
 *‘a box of plastic lids’

¹⁹ Note that the head of (30)b M-commands N3, and that N2 M-commands N3 as well. H-government would still fail in both cases. First, because neither N1 nor N2 are *sisters* of N3, and in the case of N1, there is an overt intervener which is M-commanded by N1 and M-commands N3, namely N2. The failure of either N1 or N2 to construct with N3 therefore provides evidence against replacing *sisterhood* with M-command in the H-government definition in (28). For evidence against replacing *sisterhood* with c-command, see section 9.2.

²⁰ While [[N1 N2] N3] examples are not frequent, they are productive. For instance, [[*bejt sefer*] *sade*] ‘field school’; [[*joḡv-ej roḡ*] *moḡaṣa*] (lit. head seaters of the council) ‘council chairs’, [[*tabáṣat nisuḡim*] *zahav*] ‘wedding ring of gold’, [[*ṣakrav nemalim*] *ha=jarden*] ‘Birulatus Israelensis (lit. The Jordan’s ant scorpion)’.

6.1 Is complex P a possibility?

(34) a. Very often, the Pivot is not attested as a free noun (nor had it existed in BH).
 b. And even when it is, the Content that the P_{Piv} conveys is frequently unrelated to the Content of Ns and Ps when occurring independently. Furthermore, distinct Ps with an identical Pivot may give rise to a very distinct Content.
 c. The morpho-phonology of P_{Piv} corresponds to a higher register, arguably historical, and is distinct as such from that of P+[_{DP} N].

(35) a. [PPLoc [Ploc meʔahorej] [DP [NP PLACE [DP ha=fulhan]]]]
 behind DEF=table
 b. [PPLoc [Ploc [DP [XP [X meʔahorej] [NP PLACE DP]]]]]

(36) a. [_P meʔahorej [_{DP} ha=fulhan DEF=table]]
 behind
 b. [_P [me_P-ʔahorej_N] [_{DP} ha=fulhan DEF=table]]
 behind

An important argument against the structures in (36)a-b (or for that matter (35)a,b) is that they should not allow a Construct formation between Pivot and N (N+N), as Constructs require an N head that H-governs its complement (cf. (28) and related discussion), a requirement met neither by (36)a-

²² Unlike some recent work on PPs, we do not postulate abstract variables such as PLACE, TIME, CAUSE etc. in our structure. The overwhelming richness and nuance of MH PPivs would require an extensive array of such abstract variables, thereby annulling any potential explanatory advantage. For additional comments, see fn. 42.

b nor by (35)a-b. To explain the Construct state form of the Pivot, proponents of (36)a-b or similar suggest that *all* prepositions, both simple and complex, form a Construct with the (head of) their (nominal) complement (Siloni 2001; Botwinik-Rotem and Terzi 2008; Botwinik-Rotem 2008a,b, i.a.). We note, first, that this would hardly suffice for (35)a,b, where the presumed head (P or X) is not adjacent to its N complement. In (35)b, with X postulated as the head of a specifier, c-command fails as well. More importantly, all executions which postulate a complex P head of the type in (36)a-b abstract away, summarily, from the nominal properties of the Pivot outlined in some detail in the previous subsections, and amount to attributing to prepositions a slew of nominal properties for which there is simply no evidence in simple prepositions.²³

In contrast, the present study takes at face value the nominal properties of Pivots and in particular, the fact that the Pivot acts as the *nominal* head of a Construct. This, together with the fact that the Construct occurs within a well-defined constituent, perforce blocks the formation of a constituent between P and the Pivot. Thus although {P, Pivot} do act as a unit with morpho-phonological properties and Content, they are neither a constituent nor a (reanalyzed) terminal, and a different account must be provided for their emergence as a Content and realization unit. In the remainder of this section we provide explicit arguments against the claim that PPivs are (potentially complex) projections of P.

Before turning to these arguments, it is worthwhile addressing briefly the frequent assumption that PPivs are fossilized expressions representing an older layer of the language, where they were presumably compositional and phonologically non-exceptional (e.g., Erb & Berman 2025). If, indeed, PPivs are to be traced back to some earlier period of the language, that period would need to precede Biblical Hebrew, where PPivs are already abundant (see Gesenius 1910; Hardy 2022 among others).²⁴ Even more problematic for this claim is the fact that the formation of PPiv is productive, and coinages such as *be=mahalaχ* ‘during’ (lit. ‘in (a) gear’, ‘in (a) move’) or *li=fkudat*, ‘to the order of’ in a banking context are clearly modern formations. Note that despite its modern borrowed origin, *li=fkudat* displays the morpho-phonological characteristics otherwise associated with a higher and presumably earlier register.

An even more striking example of the contemporary nature of PPivs comes from the (intransitive) PPiv *bi=djuk* (also *bi=dijuk*), ‘right, indeed’, lit. ‘in-precision’. The noun *djuk* ‘precision’ exists independently in MH, and can serve as a complement of the reposition *be*, with the result being *be-dijuk* (‘with precision’) and with *be-dijuk* allowing in this *compositional* context both the definite article and modification (*b=a=djuk ha-meravi* ‘with the maximal precision’). As for its PPiv use, it turns out to be a borrowing back into MH from Yiddish, where its meaning correlates with German *genau* ‘correct, indeed’.²⁵ The caveat is that in Yiddish, it is pronounced as *bedíjek*. Having been borrowed back into MH, it not only acquired the (contemporary) MH final stress, but also went on to adopt the *bi* allomorph of P otherwise attested either in high registers or in PPivs, and to suppress (optionally), in its PPiv use alone, the vowel following /d/ (*bi=djuk*, ‘indeed’, but **be=djuk meravi* ‘with maximal precision’). It thus emerges that the form, when integrated back into Hebrew, effectively resulted in the creation of a novel PPiv with non-compositional Content, as well as phonological characteristics unique to its PPivs occurrence. Whatever grammatical operation is responsible for the emergence of PPiv, then, it is not an archaic relic, but an existing, contemporary option.²⁶

²³ Following Siloni (2001), Botwinik-Rotem (2008a,b) postulates the existence of a phonological/prosodic word for the Constructs in (35)a,b, consisting of P/X and the nominal head of DP. That postulate, however, requires the formation of a prosodic word without adjacency between the head and the complement, clearly contrary to Siloni (2001). It also requires, at least in the case of X, a Construct formation that is not between a head and its genitive Complement, likewise rejected in Siloni (2001). For our own position on the Construct as a phonological/prosodic word, see section 9.

²⁴ Historically, PPivs have been argued to have emerged in Proto North-West Semitic, and are attested, beyond Hebrew, in Aramaic and Ugaritic. Evidence of their existence prior to that is at best scant.

²⁵ Avnion online dictionary, [בידוק פירוש: מילים נרדפות, ביטויים ועוד | מילון אבניאון](#); Elon Gilad (p.c.)

²⁶ An independent reason to reject the view of PPivs as unanalyzed P heads is the huge number of extremely nuanced prepositions that would emerge as a result. This is precisely what led Schleizer (1998) to propose that P is not a closed class in Hebrew, and to postulate a productive rule of P formation in Hebrew.

6.2 Recursion

Proceeding to our distributional comparison of simple P's and PPivs, we note first that simple P's (see list in (7)) allow P-recursion preceding the Pivot as in (38)-(39).^{27, 28}

- (38) a. *máʕal* → *mi-máʕal* → *mi=máʕal=a*
 ASCENT from=ASCENT from=ASCENT=ward
 'ascent' 'above' 'from above'
- b. *máʕal* → *le-máʕal* → *le=máʕal=a* → *mi=le=máʕal=a*
 ASCENT to=ASCENT to-ASCENT=ward from=to=ASCENT=ward
- (39) a. *táħat* *ha=mita* → *mi=táħat* *l=a=mita* → *mi=mi=táħat* *l=a=mita*
 BOTTOM DEF=bed from=BOTTOM to=DEF=bed from=from=BOTTOM to=DEF=bed
 'under the bed'; 'under the bed'; 'from under the bed'
- b. i. *ka=thila* → *le=xa=thila* → *mi=le=xa=thil-a(t)* → *ke=mi=le=xa=thil-a(t)*
 as=START to=as=START from=to=as=START-FSG(CS) as=from=to=START-FSG(CS)
 'as a start' 'in the beginning' 'a priori' 'as (if) a priori'
- ii. *b=a=thil-a* → *ke=v=a=thil-a*
 in=DEF=START-FSG as=in=DEF=START-FSG
- c. i. *le=vad* → *bi=l=vad*
 to=★ in=to=★
 'alone' 'only'
- ii. *le=vad* → *mi=l=vad*
 to=★ from=to=★
 'alone' 'except'
- d. *le=ʕum-at* → *ke=le/ki=l=ʕum-at*
 to=FSG.CS as=to=★-FSG.CS
 'juxtapose' 'in reversal of'

By contrast, PPivs never allow recursion regardless of the plausibility of the construal:

- (40) a. **me=ʕal* *mi=táħat* (*l=a=mita*)
 from=TOP from=BOTTOM (to=DEF=bed)
intended: above under the bed
- b. **li=fn-ej* *be=mé/εχ* *ha=ħag*
 to=FACE-MPL.CS in=DURATION DEF=holiday
intended: before the duration of the holiday
- c. **bi=glal* *li=krat* *ha=ħag*
 in=★ to=★ DEF=holiday
intended: because of anticipating the holiday
- d. **le=jad* *li=fn-ej* *ha=bájit*
 to=HAND to=FACE-MPL.CS DEF=house
Intended: next to the front of the house

6.3 (in)transitivity

Simple Ps in MH are always *overtly* transitive: they never occur without an overt complement (including elliptical contexts). (41)a-b are strictly ungrammatical regardless of the syllabic weight of

²⁷ See Hardy (2022) on recursive occurrences of P in Biblical Hebrew.

²⁸ The final unstressed -a in (38) is a directional particle and probably P or some form of agreement with it. It is appended to fixed locative nouns (cf. i), and is always unstressed. Logically, it scopes over the stem to its left, and does not allow a complement to its right:

i. mizráh=a	maʕaráv=a	tsafón=a	daróm=a	ha=bájt-a	yamín=a	kadím=a
east=ward	west=ward	north=ward	south=ward	DEF=home=ward	right=ward	front=ward

the preposition:²⁹

- (41) a. *nasáʕ-nu l-a=mapali ʃuv ve=ʃuv ʕad ʃe-nimʔas la-nu mi(n)---
travel-1PL to-DEF=waterfall again and=again until that fed-up to-us from ---
intended: ‘we travelled to the waterfall again and again until we were fed up with it’
b. *raʔi-nu binjan ʔadaʃ ve dán-nu be --- *ʕaʕád-nu ʔel / ʕad ---
saw-1PL building new and discuss-1PL in marched-1PL toward / until
intended: we saw a new building and we discussed it--- we marched toward it /up to it’

In contrast, PPivs can occur without a complement. This is particularly clear in (42)b-c, where the PPiv has an optional DP complement which constructs with the Pivot. When the complement is absent, the Pivot regains its non-Construct realization:

- (42) a. ha=géʕem ba me=ʕal o mi=táħat
DEF=rain came from=ABOVE or from=BELOW
‘the rain came from above or from below’
b. ha=ʕets mi=le=ʕan-im o me=ʔaħor?
DEF=tree from=to=FACE-PL or from=BACK
‘Is the tree in front or behind?’
(cf. ha=ʕec li=ʕn-ej ha=bájit o me=ʔaħor-ej ha=bajit
DEF=tree to=FACE-PL.CS DEF=house or to=BACK-PL.CS DEF=house
‘the tree is in front or behind the house’)
c. haláχ-nu mi=saviv/mi=jamin (cf. mi=sviv/mi=jmin ha=bajit)
go-1PL from=ROUND/from=RIGHT rom=ROUND/from=RIGHT DEF=house
‘we went around/on the right’ ‘we went {around/on the right of} the house’

Beyond the contextual and possibly elliptical cases in (42), PPivs freely and commonly occur without a complement, including with otherwise unattested Pivots. A small sample is in (43). No such cases are ever attested with simple Ps:

- (43) a. be=heħlet bi=χlal be=diʕavad be=ʕéʕsem
in=DECISION in=RULE in=★ in=BONE
‘definitely’ ‘altogether’ ‘a posteriori’ ‘actually’
b. la=ħalutin le=maʕréaʕ le=miʕi le=gámre
to=★.PL to=★ to=★ to=★
‘completely’ ‘in advance’ ‘cleanly’ ‘completely’
c. mi=jad mi=zman mi=le=χa=thíl-a mi=néged
from=HAND from=TIME from=to=as=BEGINNING-F from=OPPOSITE
‘immediately’ ‘long ago’ ‘a priori’ ‘across’
d. ka=ʕur-a ka=ʕet
as=LINE-F as=TIME
‘properly’ ‘now’

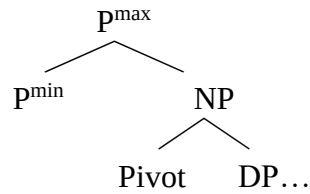
6.4 Explaining recursion and (in)transitivity

If, as we suggest, P always projects as ‘simple’ P and the Pivot heads its complement as in (44)a,b the recursion of P becomes a straightforward matter, depicted in (45):

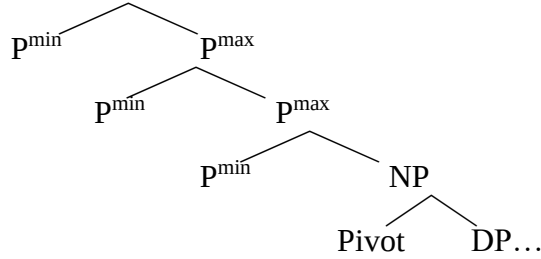
²⁹ Stranding is impossible across the board in MH either after P or following a Construct head. Absent pied piping, both simple P and Construct nominal heads sport an agreeing clitic functioning as a resumptive pronoun. As a result, no contrast is either attested or expected between PPivs and simple Ps in extraction cases.

(44) a. [$P_{\max} P_{\min}$ [D ...N ...]...]

b.

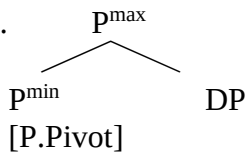


(45)

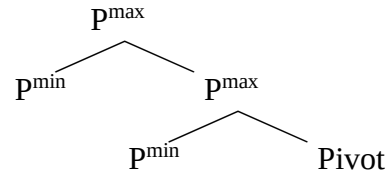


Suppose, now, that PPivs are P heads, either unanalyzed, as in (46)a, or branching, as in (46)b:

(46) a.



b.

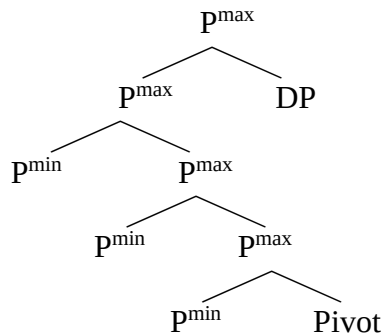


Assume now that recursion affects $P_{\min}$ in (46)a. The result would be as in (47) and is, as we already saw, ungrammatical (cf.(40)):

(47) [_PP.Piv [_P P.Piv [_P P.Piv [_{DP} ...]]]]

If, however, PPiv is complex, but still projects P, as in (46)b, one could propose that recursion affects the P-head of the P+Pivot, preceding the Pivot complement, with the hypothetical structure in (48):

(48)



The structure in (48) would allow for a coherent statement that ‘simple’ P can recur, while still holding on the view that PPiv is a constituent. That relief comes at a distinct cost, however: in (48) the DP complement is no longer a sister of any instantiation of $P_{\min}$ nor is it a sister of the Pivot. The obligatory emergence of a Construct between PPiv in its entirety and the DP, or between Pivot alone and the DP hence cannot be modelled. We therefore must reject (46)b as the structure for PPiv.

Turning to (in)transitivity, we propose the generalization in (49), according to which prepositions in MH (as in great many other languages) are inherently transitive. Even more strictly, we propose an obligatory *nominal* projection within the complementation line of P. (49) does allow a PP complement of P (recursively), providing that eventually any sequence of P^* would include a direct nominal complement:

(49) Given $P_i^{\max}$ and α the sister of $P_i^{\min}$

a. α is a projection of N; or

b. α is $P_j^{\max}$ and β , the sister of $P_j^{\min}$ is a projection of N

The absence of intransitivity for true prepositions follows directly, of course, from (49). The

endemic occurrence of ‘intransitive’ PPivs follows just as directly, but only under the assumption that the Pivot is a nominal element, and a complement of P. As such it satisfies (49). Under either structure in (46), this result cannot be derived without additional stipulations.^{30,31}

6.5 ‘Bare’ Pivots

Consider now the italic tokens in (50), where a single item precedes the DP, and with that item exhibiting clearly nominal properties such as bound CS morphology and syncope (50)c,g), plural and gender marking ((50)b,c,e), and templatic nominal morphology (50)a,b,d), suggesting at least *prima facie* that they are instances of N and not instances of P:

- (50) a. *ʕékev* ha=milhama
FOOTSTEP DEF=war
‘because of the war’
- b. *tmur-at* ha=hamlatsa
EXCHANGE-FSG.CS DEF=recommendation
‘in exchange of the recommendation’
- c. *ʔaħr-ej* ha=mesiba
AFTER-MPL.CS DEF=party
‘after the party’
- d. *térem* ha=milhama
★ DEF=war
‘before the war’
- e. *ʔod-ot* ʃlómo
★-FPL Shlomo
‘about Shlomo’
- f. *ʕavur* ʃlómo
★ Shlomo
‘for Shlomo’
- g. *sviv* ha=bájit
AROUND DEF=house
‘around the house’

As it turns out, we are now equipped with tests which can distinguish between (simple) P and PPivs, and which can therefore be applied to the tokens in (50) to determine their status.

Turning first to recursion, possible for P but not for PPivs, we find that the tokens in (50) block recursion (51), on a par with PPivs:

- (51) *Recursion:*
- a. *(be=)*térem* *ʔaħr-ej* ha=mesiba
before after DEF=party
intended: ‘before the period after the party’
- b. *(mi=)*sviv* (mi) *táħat* ha=mita
around (from) under DEF=bed
intended: ‘around [under the bed]’
- c. *(ba)*ʕavur* *tmurat* ha=hamlatsa
★ in-exchange DEF=recommendation
intended: ‘for in exchange of the recommendation’

³⁰ It follows from the logic of our discussion that in intransitive forms such as *me=ʕal*, *ʕal* is not P but a Pivot (‘top’). Similar logic holds for *ʕad*. See fn. 33 for further discussion.

³¹ We assume that P in PPivs, including recursive ones, is a P projection line headed by some syntactic terminal P (or *p*) distinct from that of its nominal complement. By the logic of the categorial system, then, such P should dominate a root and be embedded within its own functional spine. This leaves open the possibility that P is a heterogeneous syntactic category, and thus possibly functional in other contexts. We leave aside these matters. For discussion of these matters see Deacon (2017) and references cited therein.

‘Intransitivity’, however, is possible, again on a par with PPivs, but in contrast with simple Ps:

- (52) *Intransitivity (tr. Indicates optional complement):*
- a. (hitsbáʕnu) *néged* (tr.) / (be=)jáhád
(we voted) *OPPOSITE* / (in=)TOGETHER
‘we voted against/together’
 - b. (ʕaʕadnu) (mi=)saviv (tr.) / (ʕad=)halom
(we marched) (from=)AROUND / (until=)★
‘we marched around/‘thus far’
 - c. (garnu) (mi=)mul (tr.)
(we lived) (from=)★
‘we lived across’

On the basis of these tests, we conclude that the ‘bare’ nominals in (50) are Pivots within PPiv combinations in which P is zero-realized (P_∅):

- (53) [P P_∅ [N Pivot [{DP, PP, CP}]]

Not surprisingly, we find that most ‘bare’ Pivots can also occur with an overt P, frequently (but not always) with the same Conent, lending additional support for the presence of a (simple) P_∅:

- (54)
- | | | | | | | |
|----|---------------|-------------------------------|------------------|-----------------|------------------|-----------------|
| a. | <i>ʔod-ot</i> | <i>ʕal=ʔod-ot</i> | g. | <i>saviv</i> | <i>mi=saviv</i> | |
| | TALE-FPL | on=TALE-FPL | | AROUND | from-AROUND | |
| | ‘about’ | ‘about’ | | ‘around’ | ‘around’ | |
| b. | <i>ʔaħar</i> | <i>le=ʔaħar</i> | h. | <i>sviv-ot</i> | <i>be=svivot</i> | |
| | AFTER | to=AFTER | | ROUND-ABS.N | in=VICINITY | |
| | ‘after’ | ‘after’ | | ‘vicinity’ | ‘vicinity’ | |
| c. | <i>mul</i> | <i>ʔel mul</i> | mi= <i>mul</i> | i. | <i>ʕéver</i> | <i>me=ʕéver</i> |
| | ★ | to ★ | from=★ | | SIDE | from=SIDE |
| | ‘across’ | ‘across’ | ‘across’ | | ‘side’ | ‘beyond’ |
| d. | <i>térem</i> | <i>be=térem</i> | j. | <i>ʕavur</i> | <i>ba=ʕavur</i> | |
| | ★ | in=★ | | ★ | for=★ | |
| | ‘before’ | ‘before’ | | ‘for’ | ‘for’ | |
| e. | <i>néged</i> | <i>mi=néged</i> | le= <i>néged</i> | k. | <i>táħat</i> | <i>mi=táħat</i> |
| | AGAINST | from=AGAINST | to=AGAINST | | BOTTOM | from=BOTTOM |
| | ‘against’ | ‘across’ | ‘across’ | | ‘under’ | ‘under’ |
| f. | <i>nóħaħ</i> | <i>le=nóħaħ</i> ³² | l. | <i>tmur-at</i> | <i>bi=tmura</i> | |
| | ★ | to=★ | | EXCHANGE-FSG.CS | in=EXCHANGE | |
| | ‘in-view-of’ | ‘in-view-of’ | | ‘in exchange’ | ‘in exchange’ | |

Taking inventory of our results thus far, we would like to propose the following:

- (55) a. The set of prepositions (P^{min}) in Hebrew is restricted to the nine ‘simple’ prepositions in (55)b, which are bare consonants, single syllables or ∅. All are extrametrical clitics.
b. ‘Simple’ prepositions (P^{min}): *ʕim*, *ʕal*, *ʕad*, *ʔel*, *mi*, *l*, *k*, *b*, ∅

While the discussion of the comparison between simple Ps and PPivs has been relatively brief, it nonetheless lays the ground for a three-way typology: (i) phonologically realized Ps; (ii) phonologically silent P; and (ii) PPivs. For reasons of scope, a fuller pursuit of this typology cannot be undertaken in this article. We nonetheless opted to include some important additions to this discussion in the appendix, containing further structural distinctions between Ps and PPivs, as well as pursuing in greater detail the consequences of postulating a phonologically silent P. This, we hope, would allow the reader to consult that discussion without interrupting our main narrative.³³

³² While *nóħaħ* is not attested outside its role as a pivot, the participle *noħáħ* is attested, with the meaning ‘present’.

³³ The two prepositions *ʕad* and *ʕal* occur as both P and N (including Pivots) although PPivs with *ʕad* as P are rare:

7 Content

7.1 Summary and heads up

As should be clear from the preceding discussion, we will be adopt structure (5)b, repeated in (56):

(56) [P P [N/PL Pivot [DP N]]]

(56) represents structurally a number of important properties of PPivs, as summarized in (57):

- (57) a. The Pivot is the head of a Construct.
 b. The nominal headed by the Pivot is not a full DP (but rather a projection N or PL).
 c. PPiv are *not* terminals of the type P⁰.
 d. P+Pivot is *not* a constituent.

At the core of our structural approach is (57)a, for which ample evidence was presented in sections 3 and 4, and for the inevitable structural consequence of (57)a, namely the fact that PPiv is not a constituent (57)d. But if, indeed, this structural conclusion is valid, how can we model Content matching or phonological representation for the PPiv sequence?

In section 8, we develop a Span-based analysis that seeks to provide an answer to this question. Such an analysis, however, rests crucially on our approach to Content matching, or, more explicitly, the way the grammar models the association of non-functional vocabulary with conceptual meaning, Content. This section outlines this approach.³⁴

Importantly, we assume that Content, in and of itself, is not a *grammatical* object. It is not constructed by the grammar, and the grammar cannot consult it to build syntactic structure or (formal)

i.	a.	be=ʃad in=ETERNITY 'in favor' <u>P - Pivot</u>	bi=l=ʃad-ej in=to=ETERNITY.MPL.CS 'without' <u>P - Pivot</u>	(le=) ʃad-ej to ETERNITY.MPL.CS 'forever' <u>P - Pivot</u> -NP complement	ʃad ETERNITY
	b.	jaʃar ʃad forest eternity 'evergreen forest, primeval forest' N – <u>NP</u> complement	kifʔat ʃad frost.FSG.CS eternity 'permafrost' N – <u>NP</u> Complement	jarok ʃad green eternity 'evergreen' A/N – <u>NP</u> Complement	
	c.	ʃad=tom until=END 'up until the end' <u>P - Pivot</u>	ʃad=halom until=HERE 'up until here' <u>P - Pivot</u>	ʃad=kan until=HERE 'up until here', also 'no more!' <u>P - Pivot</u>	ʃad=ʃolam until=WORLD 'forever' <u>P - Pivot</u>
ii.	a.	me=ʃal ʔel ʃal from=TOP toward top 'above' 'upward' <u>P - Pivot</u> P – <u>NP complement</u> (absent ʔel PPivs and given compositionality)			
	b.	metos-ej ʃal jets.MPL.CS top 'super jets' N – <u>NP</u> complement	gibor (ha=)ʃal hero (DEF=)top 'superhero' N – <u>NP</u> Complement		
	c.	ʃal=pi on=MOUTH.MSG.CS 'according to' <u>P - Pivot</u>	ʃal=pe on=mouth 'oral' <u>P - Pivot</u>	be=ʃal=pe in=on=mouth 'oral' P – <u>P - Pivot</u>	

There's at one arguable occurrence of ʃim 'with' as Pivot, in the expression me=ʔet 'by' – ʔet being an established allomorph of ʃim.

Guided by our diagnosis, we postulate ʔetel (French *chez*, *at-poss-place*) as a Pivot with a null P. ʔetel is bisyllabic and allows at least me=ʔetel and le=ʔetel, which while possibly compositional, indicate it could be nominal. *Bejn* presents a different conundrum. In expressions such as *bejn ha=batim* 'between/among the houses' it certainly appears prepositional. However, it can also occur following prepositions (*le=bejn*; *mi=bejn*), as well as in compounds with the meaning 'inter' (*bejn leʔumi* 'international'; *bejn thumi* 'interdisciplinary'). Unlike simple Ps it can also be intransitive, as in the expression *bejn le=vejn* lit. 'between to between' with the Content 'in the interim'. Because neither ʔetel nor *bejn* ever occur as parts of PPivs, we set them aside. The genitive marker ʃel and the OM ʔet are excluded, as their prepositional categorization is debatable and they never host Pivots.

³⁴ Discussion of Content based on Borer (2013, 2014).

semantic formulas. Content, rather, resides in what we shall call *the Encyclopaedia*. By contrast with Content itself, the *labels* matched with Content units *are* (or could be) grammatical objects, and we assume that they are *categorially specified phonological representations* (potentially partial). Interfacing between the grammatical label and the encyclopaedia is a *search mechanism* (*en-search*) which takes as its input *labels* and returns Content units. By definition, en-search is successful iff a *label* returns a unit of *atomic Content*, which is to say, a unit of conceptual meaning that is not composed of the meaning of any of its independently *labelled* parts.

To illustrate the workings of the system, consider (58). The items in (58)b involve the bases in (58)a, combined with -y, in itself an English affix that merges productively with nouns to return (roughly) an adjective with the properties of the base, e.g. *silky*, *dirty* etc. We notate this relation as $y=A[N]$.³⁵

(58)	BASE (N)	Input to en-search	Content
a.	bulge, silk shade, flake sturd, flim(s) slith, mim(s)	[_N bʌldʒ], [_N sɪlk] [_N ʃeɪd], [_N fleɪk] [_N stɜːrd], [_N flɪmz] [_N slaið], [_N mɪmz]	BULGE, SILK SHADE, FLAKE no listed Content no listed Content
b.	DERIVATIVE	Input to En-search	Content
	bulgy, silky shady, flaky sturdy, flimsy slithy, mimsy	[_A bʌldʒi], [_A sɪlki] [_A ʃeɪdi], [_A fleɪki] [_A stɜːrdi], [_A flɪmzi] [_A slaiði], [_A mɪmzi]	No listed Content SHADY, FLAKY STURDY, FLIMSY No listed Content

As English -y merges with N, we will assume without further discussion that the bases in (58)a already are N. When we now attempt to match these spelled-out bases with Content, we find that some of them return (atomic, indivisible) Content (SILK, BULGE, FLAKE, SHADE), but others do not (#STURD, #FLIMS, #SLITH, #MIMS). Another round of en-searching is available, however, which attempts to match Content not with a *base*, but with a *derivative* of that base, specifically here the merged result of the base + -y. Once we attempt to match these *derivatives* with atomic Content, we find the following:

- (59) a. Some derivatives do not have an independent Content from their base (*bulgy*, *silky*). If targeted by an en-search, no Content is returned for the combined string. As a result, they can only be *compositional* – their Content must be based on the Content of their base (BULGE+y, SILK+y) with whatever relevant *formal* function is accomplished through turning it into an adjective with -y (e.g., *has the properties of N*).
- b. Other derivatives are ambiguous. They allow compositional Content based on the Content of their base (SHADE-y; FLAKE-y), but they also allow *non-compositional* Content, which entails neither FLAKE nor SHADE (*flakey*=EXCENTRIC, UNRELIABLE; *shady*=DISHONEST). We note as a matter of some significance that any individual token, say *shady*, either refers to *being in the SHADE* or *being DISHONEST*, but never simultaneously to both.
- c. The third type of derivatives, *sturdy* and *flimsy* only correspond to non-compositional Content (STURDY, FLIMSY). Their bases, *sturd* and *flims* may, of course, enter an en-search, but that en-search returns no Content. The sole Content available is for the entire derivative, which is hence, by definition, non-compositional.
- d. Finally, consider the nonce forms *mimsy* and *slithy*, from *Jabberwocky*. Here, neither the bases nor the derivatives return Content. As these are well-formed phonological and syntactic English words, their anomaly cannot be grammatical. Rather, quite simply,

³⁵ *Mimsy* and *slithy* from Lewis Carroll (1871/2010).

and like any well-formed phonological sequence, they may or may not correspond to Content, and by definition, nonce forms do not.

The conclusions that can be drawn from this illustration are as follows:

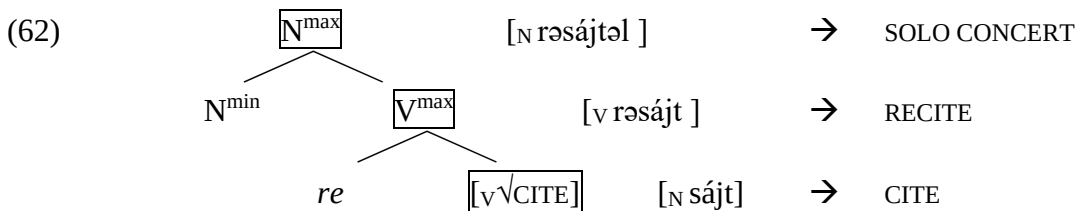
- (60) One list, one access point
- a. There exists a reservoir of (conceptual/conventionalized) atomic, indivisible Content units, call it the *encyclopedia*. Mediating between the *encyclopedia* and *grammatical* representations is a Content search engine (*en-search*), which matches *labels* with encyclopedic Content units, where by *label* we refer to a (partially) spelled out, *qualifying* representation (we return shortly to these qualifications).
 - b. *Non-Compositional Content* (=atomic Content) is the output of a *single successful en-search* and is equally associated with *cat* or with *transformation* or *naturalize* (with their listed Content). Roots may happen to be phonologically co-extensive with a qualifying string which is matched with Content (e.g., [_N cat] as opposed to $\sqrt{\text{CAT}}$) but do not, as such, have grammatically significant independent Content of any sort, nor do they represent a privileged domain for Content matching.
- (61) a. Content matching is optional (*sturd*, *silky*, *slithy* – no Content).
b. Content matching is cyclical (SHADE, SHADY).

But

- c. One Content per domain (SHAD-y or SHADY, but not both).
- d. Content, once matched, cannot be modified or elided (once SHADE cannot become SHADY)

7.2 The domain of Content

Under the assumption that Content *can* be matched with non-terminals and that morphologically hierarchical structures are syntactic, an important question concerns delimiting the syntactic *domain* of Content.³⁶ As is well-established, categorial morphology (broadly derivational morphology) may be associated with Content shifts away from the base, but not so (much of) functional morphology. As such, the properties of categorial morphology are consistent with those of the N-y combinations, which can track the meaning of the base, but need not do so, and which may be Contented even if the base is not (*sturdy*, *flimsy*). Suppose we now define the term *C-core* in reference to hierarchical structures (simple or complex) which only includes major categorial labels (so-called) A, V, N, and (some instances of) P, henceforth C-functors.³⁷ Subject to extensions to be motivated below as well as in section 8, suppose the primary domain of Content is the C-core, which is to say that en-searches are delimited by the C-core, illustrated by the range of Content matches available for (62):

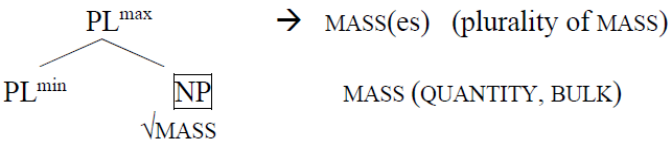
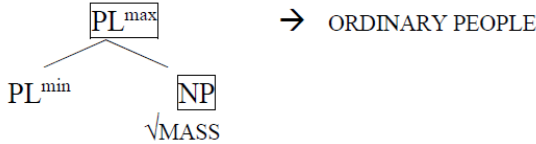


And similarly *naturalize*, *editorialize*, *civilization*, *existentialism* and others. Up to the limits otherwise allowed by the combinatorial rules of the syntax, then, any morpho-phonologically well-formed string of categorizers can serve as the input to en-searching. As a Content domain, the C-core is already structurally bigger than that proposed by Arad (2003) and Marantz (2013), which is restricted to the first categorization domain (by assumption containing only the uncategorized root

³⁶ See much contemporary discussion including at the very least Arad (2003); Borer (2012, 2014); Marantz (2013); Harley (2014) among others.

³⁷ Adverbs are set aside due to their categorially mixed nature. Recall that we assume P in PPiv is a major category (cf. fn.30).

and a categorizer). Directly below and in section 8 we will proceed to show that it is in fact even bigger. Indeed, it turns out that C-core in and of itself does not suffice to properly define the domain of *en-searches*, with English pluralia tantum serving as a prime example of its limits:³⁸

- (63) a.  → MASS(es) (plurality of MASS)
MASS (QUANTITY, BULK)
- b.  → ORDINARY PEOPLE

The input to Content can involve just [_N mæs], as in (63)a, returning the Content MASS, to which a plural marking can be compositionally added. However, the input [_{N-PL} mæsəz] is available as well, returning, for (63)b, Content which is unrelated to that of non-plural MASS, that is *masses*: COMMON PEOPLE. Similar situation holds in English *briefs*, *rapids*, *pants* and others. The participation of plural marking in Content matching contrasts, for example, with the behavior of the definite article, with English (at least) never returning Content for definite nominal expressions which is distinct from that of indefinite.³⁹ Thus while a derivational categorizer such as -y may rescue, so to speak, the string [_N sturd], by allowing it to be part of a Contentful unit, and while plural marking in interaction with its base may define a qualifying domain for en-searching, this is not the case for strings containing DEF or TENSE, with *sturded* and *the sturd* as well as *massed* and *the mass* clearly larger than the qualifying search domain for Content.

To summarize, the maximal domain allowing en-searching includes not only the C-core, but also the first functional node merging with the C-core. The relevant definitions are in (64) (and see section 8 for an extension to *Spans*):⁴⁰

- (64) a. α is a C-core iff α is (a projection of) *C-functor* and there is an intransitive β (root) such that α dominates β and for all x contained in α , x is C-functor, α dominates x , and x dominates β .

³⁸ In Marantz (2013) and Harley (2014), among others, Content is fundamentally root based, thereby crucially associated with a terminal. The role played by the categorizer, in turn, is accomplished through an allosemantic model. In that model each root constitutes a family of semes which may be individually selected in the context of a particular categorizer, or alternatively, in the context of a prefix. In such a system, therefore, Content is never associated with a non-trivial constituent. To illustrate, in such a model the root $\sqrt{\text{STURD}}$ has the Content *sturd*(y) which is selected (only) in the context of - y . By comparison, in the model put forth here we assume that Content can, and is, matched with [_A [_N $\sqrt{\text{STURD}}$] y] as a constituent as a non-terminal.

Allosemic models, we note, encounter a serious problem when it comes to forms such as *recital*. Specifically, in such cases the allosemic approach would require the roots $\sqrt{\text{CITE}}$ to be associated with an alloseme in a non-local environment (e.g. $\sqrt{\text{CITE}}$ has the alloseme SOLO CONCERT only in the context of [*re*---] *al*), or alternatively it would require the phrase [*re*[*cite*]] to have the meaning SOLO CONCERT in the context of -*al*, thereby requiring the association of Content with phrases after all. An even trickier case is presented by forms such as *conjecture*, as neither [_{CON} [_{JECT}]] nor [[_{JECT}] *ure*] have Content, which is to say that neither *con*- nor -*ure* select a seme for $\sqrt{\text{JECT}}$. Arguments for explicitly excluding an allosemic or alloseme-like Content approach to PPivs are found in section 8.2

³⁹ In other words, the definite article, even when obligatory, does not impact Content. While *the* is obligatory in expressions such as *the Bronx* or similar, the Content *Bronx* stays stable when the article is omitted, e.g. in *Bronx-lover*, showing that the obligatoriness of the article, however modelled, is not Content related.

⁴⁰ Atomic Content, we note, could be available for some [_{N+GEN+PL}] (e.g. in Hebrew *gvurot*, lit. ‘braveries=FPL’, but also ‘80 years old’) indicating that the formulation for F[C] might involve the lower functional spine, and not just the first merging F(C)-bounder. We set this matter aside.

- b. α is maximal iff α is C-functor and there is no γ such that γ is a C-functor and γ immediately dominates α .
- (65) Where α and β are Major Syntactic Categories, α is *C-functor* iff α is a terminal that projects α and defines its complement space as β ($[\alpha[\beta]]$)
(*dirty*: $[A[N]]$; *-able*: $[A[V]]$ – *readable, watchable*; *al*: $[N[V]]$ *arrival, revival* etc.)
- (66) a. Σ is *en-searchable* (Content-searchable) iff
i. Σ is a C-core; or
ii. Σ is $F[C^x]$ -bounder for C^x (and where the superscript x stands for a major categorial label)
b. $F[C]$ -bounder: the first segment of the functional spine (F) dominating the C-core

The appropriate definition of the domain of Content will play a major role in our analysis of Hebrew PPivs, to which we turn in section 8 with the result that at least some of the qualifying conditions for en-search would have to be redefined.⁴¹

7.3 The case for phonological input to en-searches

Our discussion thus far focused on the syntactic domain for Content. Evidence for a specifically *phonological* representation for (encyclopaedic) labels is available from distinct Content returned from an identical base in the context of functionally identical, but phonologically distinct C-functors. An illustration is available from English deverbal nominals. As is well known, English $N[V]$ may occur as *-ation* (with its allomorphs *-ition, -ion* etc.), as well as *-ance/ence, -ment, -al* and possibly *-age* and *-ure* as well, all of whom allow a so-called action/process nominal to emerge (other deverbal nominals such as agentives are set aside in this illustration). In a significant number of cases the source verb allows for multiple $N[V]$ realizations. When that is the case, the resulting compositional nominals (i.e., those with argument and event structure) are virtual synonyms, with differences in use mostly associated with professional registers (67). So-called R-nominals – typically with a result reading – are synonyms as well, providing they are compositional (68).

- | | | | | | |
|------|---|------|--|---|---|
| (67) | a. Kim's <i>transmission</i> of the files
b. Kim's <i>transmittance</i> of the files
c. Kim's <i>transmittal</i> of the files | (68) | a. The <i>transmission</i>
b. The <i>transmittance</i>
c. The <i>transmittal</i> | } | was successful
lasted 3 minutes
is in front of me |
|------|---|------|--|---|---|

The synonymy falls apart, however, once we consider non-compositional, atomic Content. *Transmission* can have the Content GEARBOX. *Transmittance* and *transmittal* cannot. Similarly for *recital* and *recitation*, where only the former could refer to a SOLO CONCERT. Likewise, for *committal/commitment*, only the former has the atomic Content BURIAL. Within the domain of the verb, we have pairs such as *liquidate* and *liquefy*, with *liquidate* alone having the Content BANKRUPT as well as MURDER(V). Very much along similar lines, and remembering that $F[C]$ -bounders can be part of the input to en-searching, we have the pairs *oxen* and *oxes*, with only the latter having the possible interpretation of STUBBORN PEOPLE (see in particular Acquaviva 2008 for discussion).

Crucially, in compositional contexts the relevant categorial morphemes (i.e. *-tion, -al, -ance*) are formally identical: all are instances of $N[V]$, felicitous in identical syntactic contexts and with an identical potential emergence of argument structure. As a result, we cannot appeal to the *category* of the suffixes to account for the emergence of distinct Content. Rather, it emerges that in *identical* syntactic contexts, and with formally *identical* C-functors, Content differences can only be anchored to the distinct *phonological* representation of *-tion, -ance, or -ment*. As well, without substantial supplemental assumptions, an LF Content search would fail. An allosemy-based approach to Content on the other hand (as in Marantz 2013; Harley 2014 among others) could only succeed under the assumption that there exists an alloseme of (for instance) $\sqrt{(trans)MIT}$ that means GEARBOX, that can never occur in V contexts, and that requires nominalization with *-tion* to emerge. Effectively it

⁴¹For some cross linguistic illustrations of the inclusion of $F(C)$ -bounders within the domain of Content see (81)-(85) in the context of the revised definition of the Content domain in section 8.

translates to the claim that *transmission* under its GEARBOX reading is altogether unrelated to $\sqrt{(trans)MIT}$. These reasons in and of themselves already indicate that phonological representations constitute a crucial part of the label of encyclopaedic Content units. We return to this conclusion in the context of MH PPiv in the next section.

8 Resolving the paradox

8.1 Preliminaries, and English 3-word prepositions

We began this paper by showing that PPivs present a paradox: while the P-Pivot unit has unique, atomic Content, at times contingent on unique fixed phonological representation, it is nevertheless not a constituent. But if indeed the syntax of PPivs is (broadly) as in (5)b (repeated in (69)), some modification is required in conventional views on the relationship between constituent structure and substantive meaning, specifically allowing (phonologically continuous) Content to be linked with a non-constituent. We note in this context that by making explicit reference to the root, the definition of C-core (64)a explicitly references a constituent, thereby excluding the sequence P>Pivot in (69) and similar:

(69) [P P [N/PL Pivot [DP N]]]

This notwithstanding, the availability of (atomic) Content sequences such as <P, Pivot> in (69) finds independent support by being a particularly plausible structure for (so-called) 3-word prepositions in English, whose properties are a subset of MH PPiv:⁴²

- (70)
- | | | | |
|----|--------------------------|------------------------|--------------------------|
| a. | by/(*in) dint of | at/(*by) the behest of | on/(*at) (the) behalf of |
| b. | by/(*in) virtue of | in/(*by) spite of | on/(*in) account of |
| c. | by/(*in) means of | in/(*by) terms of | by/(*in) way(*s) of |
| d. | in/(*at)/(*on) charge of | in/(*on) front of | in/(*on) charge of |

The following is a list of properties which English 3-word prepositions share with MH:

- (71)
- The choice of P is arbitrary, even for transparent Pivots;
 - Pivots could be otherwise unattested items (70)a;
 - Pivots could be otherwise attested items with a different Content (70)b;
 - Pivot can be singular or plural, but marking is fixed and crucial for non-compositional Content to emerge (70)c; (72).
 - Pivot is non-referential and cannot be modified (pseudo-incorporated profile) (72).
- (72)
- *In light of* the guidelines which was very bright
 - *with definite virtue of* the results

We note that while a string such as <*in light of*> has atomic Content and the realization of P is strict (and arbitrary), the claim that it constitutes a complex P head is highly implausible. In addition to the absence of any morphological evidence for such a formation, a complex P head should be capable of assigning Case to its complement directly, making the occurrence of *of* superfluous. That *of* occurs thereby suggests that the English Pivot is an independently projecting N.⁴³

Similarities aside, obscure Pivots as in (70)a,b, which is to say otherwise unattested or with Content contextually changed, are not as common in English and most Pivots track their otherwise attested

⁴² We focus primarily on transitive Pivots in English (occurring with *of*) to create a minimal comparison with the Hebrew cases, where transitive Pivots were chosen because of their crucial role Construct formation plays in our argumentation. However, 3-word prepositions in English with a PP complement are extremely common, whereas in Hebrew for reasons we do not understand they are relatively rare. For a (clearly incomplete) English corpus of the English construction, see e.g. [Three-Word Complex Prepositions - Prospero English](#).

⁴³ An intriguing question concerns English cases such as *behind* or *below*, where, possibly, a complex prepositional head *has* been formed, thereby obviating the need for *of*. By the same logic, it is clear that no such reanalysis took place for *in front of*, or by *dint of* or indeed *on behalf of*. We are grateful to [redacted] for pointing out to us the significance of such cases.

nominal meaning, or one of its polysemes.⁴⁴ Finally, in the absence of the Construct in English, the structures under consideration do not exhibit any locality-conditioned allomorphy in N+N contexts, and hence the impact on the relationship between the Pivot and the DP complement cannot be assessed. As a result, the properties of English 3-word prepositions (presumed PPivs) emerge as a subset of those attested in Hebrew, with differences primarily resulting from the periphrastic nature of the latter, vs. the presence of a (semi) synthetic N-N form in Hebrew:⁴⁵

(73)	<u>English</u>	<u>MH</u>
PR for PPiv	no	yes
atomic Content for PPiv	sometimes	endemic
[Pivot+DP] constituent	yes	yes

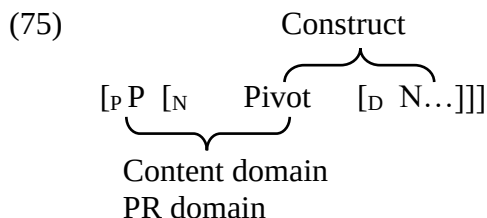
8.2 Spans

The structure we adopt in (69) has, recall, the properties in (74). The diagram in (75) repeats our paradox.

- (74) a. DP is the complement of Pivot.
b. Pivot + DP is a constituent.

Therefore

- c. The Pivot cannot form a constituent with P



If our structures and reasoning thus far are on the right track, and if, indeed, both the Content domain and the PR domain need not be terminals or constituents, then clearly a different system is required to model PR and Content, all the more so as we have already shown that the appropriate phonological representation is crucial for the emergence of the relevant Content (see subsection 2.3).

As it turns out, in the last 15 years significant research has developed allowing the modelling of properties that may hold of strings that are, in themselves, neither single terminals (possibly derived by head movement) nor constituents. Particularly noteworthy is work on *Spans*, as in Svenonius (2012, 2016), Adger (2013), and Merchant (2015), inter alia. A large body of work with a similar approach to terminals and constituents has been produced within the Nano-syntax approach (Baunaz et al. 2018 and references cited therein). However, what marks all this existing work is that it restricts its attention to the functional domain. Its focus is either phonological exponents which cross functional constituent boundaries (for instance, French P-DET forms such as *aux*), or, as in the case

⁴⁴ 9 out of the 43 3-word prepositions in the corpus we used are labelled ‘obscure’: *by dint of*, *by virtue of*, *on account of*, *on behalf of*, *in spite of*, *with respect to*, *in line with*, and with *as well as* and *in lieu of* potentially added. *At the behest of* is not included, possibly because of the definite article.

⁴⁵ Architecturally, (69) bears important similarities to the structure proposed for 3-word prepositions in English and in French in Svenonius (2010) and in Roy and Svenonius (2009). That account, however, assumes the terminal we dub Pivot is a *functional* semantic constant which may range over variables (PATH, PLACE, AXIAL PART, CAUSE etc.). The broad range of meanings associated with Hebrew PPiv’s (as well as in English), however, casts a serious doubt on the explanatory value of such semantic constants. First, Hebrew PPivs would require a vast expansion of the available inventory of formal constants, casting doubt on the explanatory value of the exercise. Second, while, for example, *be=ʕetj-o* ‘because, negative’, *bi-zxut* ‘because, virtue’, *ʕal smax* ‘based on’, *mi-pʔat* ‘because’ would all presumably involve CAUSE, they also convey additional Content to CAUSE. Unlike, for example, AXIAL PART or PATH, this additional Content could be quite nuanced. A representation that postulates CAUSE would therefore require an additional mechanism, presumably through modification, to provide the additional Content. That additional modifier, furthermore, would need to take into account not just the Pivot, but also the specific P available, and combine them to form a modification Content suitable in the context. Barring actual evidence that such an account would be syntactically or semantically simpler, we have opted not to use semantic constants in our analysis.

of Nano-syntax, the decomposition of grammatical formatives (e.g., Case markers) into distinct (non-morphemic) features to allow the creation of strings of featural heads that correspond to sets of grammatical properties with a single phonological representation. The only approaches we are aware of to the modelling of Content without terminals or constituent structure are in Borer (2021) in the context of Semitic root/template structures, and Preminger (2021), and both are theoretically sketchy and incomplete.

In what follows, we will strive to adhere to the theoretical essence of the Content model already outlined in section 7, with two important differences. The constituent presupposition in (64)a will be eliminated, and the requirement of PR for the Content domain will be introduced. In defining the notion Span we will follow, largely, the approach developed by Svenonius (op. cit.). At the core of Svenonius' approach lies the claim that item-specific phonology can be associated with ω , where ω is a *Span*, and where a Span consists of a well-defined sequence of adjacent heads. With a few modifications, we would now like to suggest that PPiv is a Span in that same sense. We will refer to that Span as ϖ ('pomega'), to highlight its distinctness from Svenonius' execution. ϖ constitutes a *phonological sequence delimited by syntactic structure*. Crucially, and beyond previous approaches to Spans, ϖ defines an en-searching domain. In (78) the crucial role of C-functores and F[C]-bounders is maintained, but ϖ as defined in (76) now explicitly allows the en-search of non-constituents and requires a phonological representation for en-searching to proceed:

- (76) ϖ_{DEF}
- Given an exhaustive sequence of heads $\Sigma \langle t_i, \dots, t_n \rangle$ s.t. for any t_k in Σ , $k \neq n$, t_k *H-governs* t_{k+1} , Σ is ϖ iff Σ has a phonological representation.
 - Σ has a phonological representation iff there is t_k , t_k in Σ , s.t. t_k has a phonological representation.
- (77) $H\text{-Government}_{\text{DEF}}$
 $\alpha^{\min}$ *H-governs* $\beta^{\min}$ iff $\alpha^{\min}$ is a sister of $\beta^{\max}$ and there is no χ s.t. $\alpha^{\min}$ c-commands χ and χ c-commands $\beta^{\min}$ and χ is phonologically overt.⁴⁶
- (78) Σ is *en-searchable* iff:
- Σ is ϖ
 - For any t_k in Σ , $t_k = \text{C}$ or F[C]-binder

Some en-searchable strings are in (79). Some *unsearchable* ones are in (80):

- (79) En-searchable iff ϖ (surface string order)
- | | | | | |
|---------|-----------|---------------|---------------------------------------|--|
| a. P+N | b. N+N | c. N+PL, N+DU | d. N ₁ +PL ₁ +N | e. N ₁ +N ₂ +PL ₂ |
| f. CL+N | g. N+CL+N | h. P+N+PL | i. ASP+V | j. V+P |
- (80) Not en-searchable
- | | | | |
|--------------|------------|-----------|---|
| a. DEF+NUM+N | b. T+ASP+V | c. [x Ø] | (where Ø is in reference to phonological realization) |
|--------------|------------|-----------|---|

Some concrete examples of Contented Spans are in (81)-(85) (F: F[C]-binder):

⁴⁶ As already noted in fn. 18, the prohibition on phonologically overt intervener in (77) is designed to block ϖ across a phonologically realized specifier. For the treatment of potentially intervening specifiers within a Span-based approach, see, in particular, Svenonius (op. cit.) and Adger (2013). An alternative formulation would require $\alpha^{\min}$ and $\beta^{\min}$ to be *phonologically adjacent*. An account cast (for instance) in terms of c-command and phonological adjacency must, in turn, deal with the potential presence of intervening phonologically null functional heads, and the extent to which they block the formation of a Span. Here, we assume without further discussion that structures such as [DEF>N] or [T>V] are semantically uninterpretable, and that for semantic interpretability, a DP must contain a NUM/Q node (at the very least) and a TP must contain ASP, even if these are not phonologically realized. Consequently, T never H-governs V and D never H-governs N, and thus sequences such as T-V or D-N are never en-searchable. We return to the question of sisterhood vs. c-command in section 9.

- (81) a. P-N le=*régel*
to=*LEG*
'on the occasion of' (MH PPivs)
b. N-N gas=light
bulk=head
jabber=wacky (EN compounds)
- (82) F-N
a. F:PL mass-es (EN)
b. F:DU mispar-*ájim* (MH)
number-DU
'scissors'
c. F:DIM bloeme-tje (Dutch, De Belder et al 2015)
flower-DIM
'bouquet of flowers'
d. F:CL tienwoe ki / tienwoe tung (Taiwanese)⁴⁷
telephone long telephone through
'telephone wire' 'telephone call'
'telephone' (instrument)
- (83) N+[N+F]
a. F:PL ʔiʃ ʔeʃkol-**ot** (MH)
man fruit.cluster-PL
'polymath'
b. F:DIM [[vlagget-jes] dag] (Dutch)⁴⁸
flag-DIM day
'first day of fresh herring'
- (84) P+[N+F]
a. F:PL ʒal=ʔod-**ót** (MH)
o= ʔ-FPL
'about'
b. F:of by dint of (EN)
- (85) PERF+V (Polish, Lazarczyk 2010)
czytała → READ
przeczytała → READ-prf
od-czytała → PRESENT-prf
roz-czytała → DECODE-prf
w-czytała → UPLOAD-prf

As used here, ϖ is a well-defined formal object, with a clear articulation of the locality conditions that govern its occurrence. Its definition, therefore, is fully falsifiable, and indeed, it could be falsified, for example, by a veritable atomic Content unit associated with DEF-NUM-N, by T-(ASP)-V, or for that matter by DEF-NUM. Similarly, it would be falsified by a discontinuous string, skipping a head (overt or covert) that is excluded from the en-search (e.g. P1-P2-N with P1-N forming a ϖ which skips P2

⁴⁷ Z. Wu, personal communication.

⁴⁸ A. Neeleman, personal communication.

(overt or covert) and with the en-search input consisting of P1-N with a variable P2.^{49,50}

8.3 Returning to PPivs

With the definitions in (76)-(78) in place, we return to MH PPivs to articulate the details of our analysis. In (86), we have a fully compositional PP, in which the preposition *le* ‘to’ merges with the complement DP that is headed by *kavod* ‘honor, respect’. *kavod*, in turn, constructs with the head of its own complement, *ha=mišpaħa* ‘the family’. Depending on register, both realizations in (86) are licit. Concretely, both the preposition and the N head have allomorphs. *le* alternates with *li*, and *kavod* alternates with χ vod or *kvod*, with the allomorphs *li* and χ vod associated with a higher register. Register notwithstanding, the choice between these allomorphs does not impact Content, and (86)a and (86)b have an identical syntax and an identical interpretation:

- (86) a. $li^H=$ χ vod^H *ha=mišpaħa* *Compositional*
 P [DP1 N.CS ... [DP]]
 ‘to’ ‘honor’ ‘the family’
- b. *le=* *kavod/kvod* *ha=mišpaħa* *Compositional*
 P [DP1 N.CS ... [DP]]
 ‘to’ ‘honor’ ‘the family’

The corresponding PPiv structure is in (87) and here, the allomorphs of P and the Pivot respectively *must* be *li* and χ vod:

- (87)

$li=$	χ vod.CS
P	[N/PL N.CS ...]
Σ , ϖ_{26} , en-searchable	

 [DP DP]
 Atomic Content unit available for ϖ_{26} $\rightarrow$ ANTICIPATING

Given our Span-based definition of the Content domain, both $li^H=\chi$ vod^H and *le=kavod/kvod* in (86)a,b are well formed Spans, but they are not en-searchable because χ vod/*kvod*/*kavod* are embedded

⁴⁹ Not excluded, however, is ϖ consisting of P2-P1-N as in (i.a), or for that matter P1 N where P1 is $\emptyset$ as in (i.b), see section 6 for discussion. Recall that for ϖ to be licit, as per (76), it is sufficient for one element in the Span Σ to have a phonological representation.

- i. a. *mi=l=vad* *ha=kir*
 from=to=★ DEF=wall
 ‘except for the wall’
 b. $P_{\emptyset}=fekev$ *ha=milħama*
 FOOTSTEP DEF=war
 ‘because of the war’

⁵⁰ The domain of Content as defined here excludes intervening functional nodes which are not F-bounders, as well as discontinuous expressions. As a result non-continuous sequences corresponding to atomic Content, such as *kick the bucket* or *toss out the baby with the bathwater* could not serve as a single input to an en-search and another Content mechanism would be required for them. We take this to be an advantageous result. The system as it stands also excludes, as a domain for a single en-search, verb-particle constructions which may be a disadvantage. Specifically, when it comes to Atomic Content, we perceive, broadly, the following continuum, with || indicating the boundary between what can or cannot be included in a single ϖ :

i. Atomic Content for:

morpheme	<derivative	<compound	<C sequence+F-bounder	<Span	(potentially ϖ)
cat	recital	walkman	briefs		
		blue moon	<i>roz-czytała</i>	<i>be=gin</i>	
			PRF=write	in=★	
			‘decode’ Polish	‘because’ Hebrew	

separable affixes/particles	<light verbs	<rigid idioms	<semi-compositional idioms	(cannot be ϖ)
give (something) up	take a shower	spill the beans	throw out the baby with the bathwater	

We leave it to future research to determine whether this partition is ultimately the most explanatory one, or whether definitions would need to be modified so as allow in a sufficiently restricted manner the search for Content to apply to some discontinuous elements.

within a full DP, and as a result, any Σ which contains both the preposition and the N would also contain the intervening nodes D and NUM/Q. Not only is H-government not met, but the emerging Span has at least two instances of intervening F, of which one, D, is most clearly *not* an F[C] bouncer. The result is that ϖ , the input to en-searching, never forms and the unit as a whole cannot be matched with Content. Content, to the extent available, must be associated separately with $\chi vod/kvod/kavod$ and with the preposition *le*, with compositionality resulting.

Not so for the representation in (87). We argued in some detail that Pivots are, at most, instances of N/PL, and specifically that the Pivot nominal is not a full DP. Σ in such cases consists either of the string P+N or of the string P+PL+N, within a configuration that meets H-government for all members of Σ and which contains at most one F[C] bouncer: PL marking (and recall that we assume that P here is a C-functor, see fn. 31). Given the phonological representation of both P and Pivot, Σ is ϖ , and can enter en-searching as in (87) to return a single atomic Content unit.

We note, finally, that an allosemy-like Pivot-in-context approach to the Content of PPivs will fail. Under such an approach Content is not matched with the PPiv as a whole, but rather, only with the Pivot in the context of a specific preposition. Under a contextual allosemic account a form such as *peʔa* would constitute a family of semes, including at the very least WIG/SIDEBURN/EDGE/CAUSE. While the allosemes WIG/SIDEBURN/EDGE would require no context (apart from pragmatics), the alloseme CAUSE would only be available in the context of *mi=* thus giving rise to BECAUSE not together with *mi=* but in its context (*mi=pʔat*, BECAUSE).

What is at stake here, we note, are not only the specific Content properties of PPivs and Pivots, but more broadly, the availability, across the board, of Content for with non-terminals, including non-constituents such as Spans.

As the reader may recall, the Pivot is not a root, and is frequently a complex derived nominal. In the PPiv *le=ʔsmitut*, ‘forever’, *ʔsmitut* lit. ‘serfdom’ is derived from *ʔsamt* ‘serf’ and in and of itself, only has the meaning composed of it (i.e. *ʔsamt* + NOM). Only in the context of the preposition *le=* does the Content ‘forever’ emerge. One could propose nevertheless an allosemy-like account, that would sever semes from roots but would still associate them exclusively with some qualifying ‘word’ units, under the assumption that the Pivot (including *ʔsmitut*) is such a terminal, but not so the PPiv. Under such an account *ʔsmitut*, as a whole, would have an atomic alloseme ‘forever’ that would require the preposition *le=* to be available (this, in actuality, would be the *only* seme of *ʔsmitut* as the ‘serfdom’ meaning is composed of the Content of *ʔsamt* ‘serf’.)

Paradoxically enough, to reach this state of affairs requires *ʔsmitut*, a syntactically complex form, to have Content as a whole. The only other option would be to assume that the Content ‘forever’ is available as an alloseme of *ʔsamt* (alongside ‘serf’) in the context [*le=[[ʔsmit] ut]*], making the alloseme selection non-local (see fn. 38 for some comments on English derivatives and locality in this context).

It goes without saying that any empirical evidence against contextual allosemy, including for derived forms, can only serve to show its limitations, but not to exclude it across the board. In what follows we therefore present positive evidence that atomic Content for the entire PPiv is inevitable.

Consider, specifically, the triplet in (88) (see subsection 6.2 for additional examples of P recursion):

(88)	<i>le=vad</i>	<i>bi=l=vad</i>	<i>mi=l=vad</i>
	to=★	in=to=★	from=to=★
	‘alone’	‘only’	‘except’

The Pivot in (88) is *vad* (or possibly *bad*) which is otherwise not attested in the language.⁵¹ In a contextual allosemic model, *vad* would nevertheless be associated with (at least) 3 allosemes – ALONE, ONLY, EXCEPT. The alloseme ALONE could only be inserted in the context of *le=*; the alloseme ONLY would be inserted in the context *bi=l=* and finally the alloseme EXCEPT would be inserted in the context *mi=l=*. In objecting to this execution we note that (a) the context for the insertion of ONLY

⁵¹ One exception is the idiomatic fixed expression *bad be=vad*, ‘simultaneously’ already an obscure semantic isolate in Biblical Hebrew.

and EXCEPT is not local. (b) Both *levad* and *milvad* occur already in Biblical Hebrew, while *bilvad* is a late addition (circa 200ad). Given the earlier existence of *levad*, as well as the shared semantic field, the most natural account for the emergence of the PPiv *bilvad* is as a meaning extension based on *levad*, which is to say a related, but non-compositional new PPiv. Deriving it, at that later stage of the language, from an otherwise meaningless, unattested *vad* makes little sense. But if, as seems highly likely, *bi=l=vad* emerges as an extension of *levad*, then at least at that historical point, *levad* is clearly associated with atomic Content as a *complete* PPiv. (c) There are several occurrences in Biblical Hebrew (as well as in Modern Hebrew) of the combination *levad mi-* lit. ‘alone from’, but in actuality, EXCEPT. Regardless of whether the composite form *milvad* has emerged from the incorporation of such a P into the P component of PPiv, or has emerged independently, it is clear that these facts can only make sense in the context of an existing, Contented *levad* thereby indicating that a single unified Content for PPivs is perforce available.

8.4 Structure, realization, Content

Two points are worth highlighting in this final subsection of our proposal. First, although the syntactic domain of Content and specifically H-government sufficed to exclude the strings in (86)a-b from en-searching, constrained syntactic hierarchy on its own does not suffice to delimit the domain of en-searching. To see this, compare (87) (repeated as (89)a) with (89)b, under the assumption that what merges with P in both is N/PL. More specifically, there is no inherent phonological or syntactic reason to *exclude* the allomorphs *le* and *kavod* from the structure in (89)b, nor is there any particular reason to assume that the result is ill-formed syntactically in (89)b but licit in (89)a. Furthermore, as both $\Sigma(le, kvod.CS)$ in (89)b and $\Sigma(li, \chi vod.CS)$ in (89)a include phonological representations, both are licit instances of $\mathfrak{w}$.

It remains the fact, nevertheless, that the $\mathfrak{w}$ corresponding to $\Sigma(li, \chi vod.CS)$ does have corresponding atomic Content, but not so the $\mathfrak{w}$ corresponding to the $\Sigma(le, kvod.CS)$. The difference is not structural. Rather, it emerges directly from the particular phonological realization of the allomorphs chosen. The string of allomorphs in $\Sigma(li, \chi vod.CS)$ *does* correspond to atomic Content. The string $\Sigma(le, kvod.CS)$ *does not*, and it is therefore ‘doomed’ to compositionality.⁵²

- (89) a. $[_P li=[_N \chi vod.CS [_{DP} N]]]$
 $\Sigma, \mathfrak{w}_{26}$, en-searchable
 Atomic Content unit available for $\mathfrak{w}_{26}$
 $\rightarrow$ *both compositional and atomic*
- b. $[_P le=[_N k(a)vod.CS [_{DP} N]]]$
 $\Sigma, \mathfrak{w}_{65}$, en-searchable
 Atomic Content *not* available for $\mathfrak{w}_{65}$
 $\rightarrow$ *compositional only*

A slightly different case, but with a similar conclusion, is in (90). The allomorph chosen for ‘honor’ is *χvod*, the very same as that chosen when it is a Pivot. The P is distinct, but it, too, occurs as an allomorph typically associated with higher register when compositional (*bi* rather than *be*), just like *li*, and is often the chosen allomorph for PPivs (for example *bi=djuk*, ‘indeed, correct’). The result is nevertheless compositional (and high), because unlike the $\mathfrak{w} li=\chi vod$, the $\mathfrak{w} bi=\chi vod$ does not happen to correspond to an atomic Content unit:

- (90) $[_P bi^H=[_N \chi vod.CS^H [_{DP} N]]]$
 $\Sigma, \mathfrak{w}_{34}$, en-searchable
 Atomic Content unit *not* available for $\mathfrak{w}_{34}$
 $\rightarrow$ *compositional only* (‘in honor of the family’)

⁵² In continuation to the discussion of allosemy in section 8.3, we note that postulating a context for the emergence of ANTICIPATING as an alloseme for *χvod.CS* would not only require the context of the preposition *l-*, but a particular allomorph of it, *li*. Therein lies, independently, strong evidence for establishing phonological contingency for Content searches.

That $li=\chi vod$ returns atomic Content, but not so $le=k vod$ or $bi=\chi vod$ is now reminiscent of the Atomic Content returned for *transmission*, but not for *transmittal* or *transmittance* discussed in subsection 7.3 (see (67)-(68)). Altogether, the correlation between atomic Content and arbitrary phonological realizations provides extraordinarily strong support to the claim that the input to en-searching must contain phonological information.

The second point worth highlighting concerns the optionality of Content matching. As just noted, e.g. $le=kavod/le=k vod$ or $bi=\chi vod$ are not ambiguous, and can only correspond to a compositional construals. But register aside, $li=\chi vod$ is perfectly ambiguous (cf. (86)). The en-search could select χvod and assign Content to it (perforce HONOR) and when it subsequently merges with $li=$ it is too late and compositionality emerges. On the other hand, the en-search could ignore χvod and opt instead to take $li=\chi vod$ as its domain, thereby returning ANTICIPATING. Such Content optionality is even more obvious in cases where neither P nor Pivot allow allomorphs, and where the choice between compositional vs. atomic Content for the P-N combination is altogether unmarked, as illustrated in (91)-(92). The reader may recall that an identical ambiguity between atomic Content and compositional construal was associated with tokens such as *shady* and *flakey* (cf. (58) and related discussion) whose analysis already required the assumption that Content matching is optional.

- (91) a. $mi=bhin-at$ $ha=kélev$
 from=*EXAMINATION-FSG.CS* DEF=dog
 ‘from the dog’s perspective’
 b. (lo jatsa klum) $mi=bhin-at$ $ha=matemátika$
 (NEG come.out nothing) from=*examination-FSG.CS* DEF=mathematics
 ‘(nothing came out) of the mathematics exam.’
- (92) a. (hitsbáf-nu) $le=máfán$ $ha=miflaga$ $ha=komunistit$
 (vote-1PL) to=*ADDRESS* DEF=party DEF=communist
 ‘we voted for the communist party’
 b. (jaláh-nu et $ha=mi\gamma tav$) $le=máfán$ $ha=miflaga$ $ha=komunistit$
 (sent-1PL ACC DEF=letter) to=address/residence DEF=party DEF=communist
 ‘we sent the letter to the address/residence of the communist party’

Having concluded the presentation of our empirical data and the argumentation for a Span-based approach to Content, we now turn to exclude a plausible, but nonetheless unworkable alternative, according to which the relationship between P and the Pivot is best handled through incorporation.

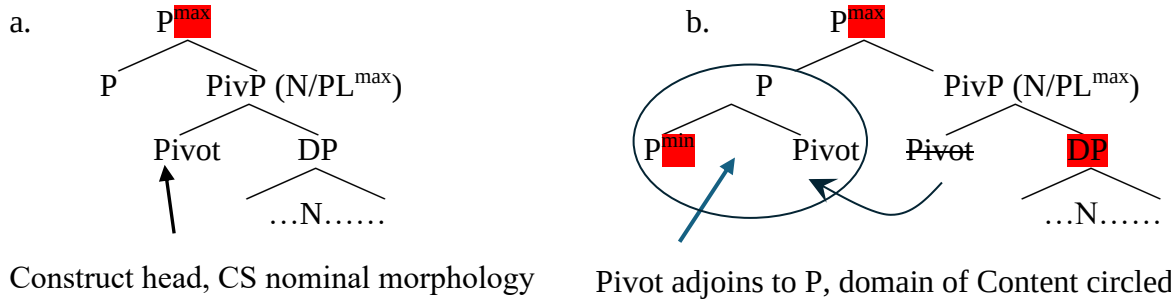
9. Against incorporation: an argument from prosody

9.1 Incorporation and the syntax of Construct nominals

The advantage of our proposed structure is that it captures syntactically the nominal nature of the Pivot, otherwise apparent in its morphology. The price, as we already noted, is the non-isomorphism between Content and constituent structure. One could argue, however, that the empirical facts, as observed and described, could be explained without assuming a complex P, while still maintaining isomorphism between constituent structure and Content, thereby making (non-constituent based) Spans unnecessary. In such an alternative account, the Pivot would still merge as the nominal head of the Construct (itself a complement of P) (93)a, but it would then *incorporate* into the preceding P as in (93)b. According to this analysis, then, the string P=Pivot would not start off as a constituent, but it would become one following *incorporation*. While there are no cases we know of in which the head of the Construct may move away without its complement, one could presumably suggest that it is precisely the truncated, non-DP N/PL nature of the Pivot that would allow such incorporation in PPivs.

In such an incorporation-based account, by assumption, the nominal morphology of the Pivot would be marked at external merge (directly or through features), but the domain of Content would be calculated on the basis of internal merge.

(93) Incorporation:



Despite the prima facie allure of this alternative, closer investigation reveals that it is incompatible with the properties of the Construct and specifically with the requirement of H-government repeated here:

(94) H-Government_{DEF}

$\alpha^{\min}$ H-governs $\beta^{\min}$ iff $\alpha^{\min}$ is a sister of $\beta^{\max}$ and there is no χ s.t. $\alpha^{\min}$ c-commands χ and χ c-commands $\beta^{\min}$ and χ is phonologically overt.

In (93)b, the Pivot does not H-Govern its complement, therefore blocking the formation of a Construct and yielding the wrong predictions concerning the morphology of the Pivot.

Two potential responses to this argument could be raised by advocates of incorporation. According to the first, the Pivot-DP would meet H-Government *prior* to incorporation thereby allowing the Pivot to acquire its bound features in a licit structural environment. It then incorporates into P, thereby allowing a P-Pivot constituent to form which serves as the input to en-searching.⁵³ As it turns out, however, prosodic factors mandate a rejection of this execution as well.

According to a second potential response by advocates of incorporation, *sisterhood* should be replaced by the weaker c-command condition in the definition of H-government. Under the assumption that the moved adjunct c-commands its remnant, the incorporated Pivot *could* thus H-govern its complement DP thereby allowing Construct formation after incorporation. As we shall see, however, there is direct evidence from the prosodic structure of Constructs that it is precisely *sisterhood* and not c-command that must be in place in Construct formation.⁵⁴

Accordingly, we now turn to a discussion of prosody in general, and the prosody of the Construct in particular.

9.2 The prosody of the Construct (including Pivot-headed)

In the past decades, multiple studies of the Semitic Construct have claimed that an explanation for its properties, including those which appear syntactic in nature, must reside with the fact that it constitutes a prosodic unit of some sort (Benmamoun 2000; Siloni 2001; Ben-Meir 2024 among others).⁵⁵ That said, few other than Faust & Torres-Tamarit (2024, henceforth F&T) formalize this claim within a phonological theory of prosodic structure. Once such a formal account is available, it becomes clear that it is entirely incompatible with an incorporation account for the formation of PPiv and therefore mandates its rejection.

Focusing on MH, F&T note that Construct aside, every word involves a single prosodic foot. In the common case of stress-final words, that foot is iambic: it includes the stressed vowel and the preceding vowel (95)a (feet are in parentheses, stress marked with an acute accent). Any syllable

⁵³ Alternatively, some phase-based account may assign spellout to the head before incorporation but still allow the head with its bound morphology to move away. We will proceed to argue directly that the prosodic nature of the Construct is such that it has to be associated with a licit *overt* representation. That excludes both a post-spellout incorporation account and a phase-spellout account.

⁵⁴ This second alternative has been proposed to us by an anonymous reviewer.

⁵⁵ While earlier accounts, from Prince (1975) onwards, recognized the prosodic properties of the Construct, the assumption was that they derive from the syntactic structure, rather than the other way around.

preceding that foot is unfooted.⁵⁶ When suffixes are added, stress generally moves to the vowel of the suffix, and so does the footing (95)b. As shown, the stress shift leaves the vowel that was pretonic in the unsuffixed base outside the foot in the suffixed form. Some lexically-marked /a/ vowels in open syllables – underlined in (95)b and subsequent discussion – syncope in this configuration (syncope is marked with a barred /a/). Other /a/s do not syncope and are by assumption lexically marked as ‘strong’.⁵⁷

(95)	<u>SG</u>		<u>PL</u>		
a.	(pakíd)	‘clerk’	b.	pā(kid-ím)	‘clerks’
	ki(farón)	‘talent’		kiḟā(ron-ót)	‘talents’
	ka(lanít)	‘anemone’		kala(nij-ót)	‘anemones’
					syncope
					syncope
					non-syncope

F&T interpret this as a prosodic effect. The weak /a/s are susceptible to syncope in open syllables, when they are excluded from the single foot of the word. Strong /a/s, however, are maintained regardless of footing.

In Constructs, as noted in section 1, main stress is carried by the syntactic non-head, and the syntactic head bears secondary stress, for example *bakbùk kóla* ‘Coke bottle’. F&T note that /a/-syncope applies to the same /a/s when the nouns they occur in are prosodic non-heads in an N+N Construct sequence (96)a–c, even though the nouns are not suffixed. In addition, it applies in cases like (96)d–g, that is, to many /a/s which, when the noun is in isolation, are pretonic and therefore maintained. Note also, for further reference, another typical change in this position: a noun which in isolation carries a stressed sequence *áj* sees that sequence reduced to *èj* in this prosodic non-head position (96)h,i (this occurs in suffixed forms as well, for instance *šejn-ájim* ‘eyes’).

(96)	<u>N</u>		<u>N+N</u>	
a.	pakíd	‘clerk’	pakíd ḥadaḟót	‘news clerk’
b.	kiḟarón	‘talent’	kiḟarón hikúj	‘imitation talent’
c.	pan-ím	‘face-pl’ ⁵⁸	pan-èj kélev	‘dogface’
d.	makóm	‘place’	makóm mifgáf	‘meeting place’
e.	matar-ót	‘targets’	matar-ót šéts	‘wooden targets (wood targets)’
f.	ḥadaḟ-ót	‘news’	ḥadaḟ-ót bóker	‘morning news’
g.	toḟav-ím	‘residents’	toḟav-èj sín	‘residents of China’
h.	šájin	‘eye’	šèjn néts	‘eagle eye’
i.	tsájid	‘hunt’	tsèjd letaʔót	‘liazrd hunt’

The syncope in (96)a–g cannot be attributed to the unfooted status of the vowel: as shown, all prosodic non-heads in (96) carry secondary stress, so they must be footed. Therefore, F&T refine their proposal. Constructs are composed of two (or more) *minimal* prosodic words (MinWd), each with its own single foot. These MinWds are grouped into one *maximal* prosodic word (MaxWd). The foot of the last MinWd is the *head foot*, and its head syllable receives main stress. The foot of a preceding MinWd (the *dependent* MinWd) only receives secondary stress. /a/s are protected from syncope only in the head foot, thereby giving rise to a syncope (for weak /a/) in the leftmost member of the Construct, that is, the nominal which is otherwise the syntactic head of the Construct.

The diagrams in (97) illustrate the structure. Prosodic Heads are signaled by straight lines and dependents by oblique lines. In (97)a, the Construct *pkid ḥadaḟót* ‘news clerk’ involves two weak /a/s, one before each stressed vowels, cf. (96)a,e. But only the first one (barred and underlined), which occupies the dependent position of the dependent foot, is syncope. The one immediately before the main stress is not syncope, because it is situated in the head foot, that is, the foot of the noun

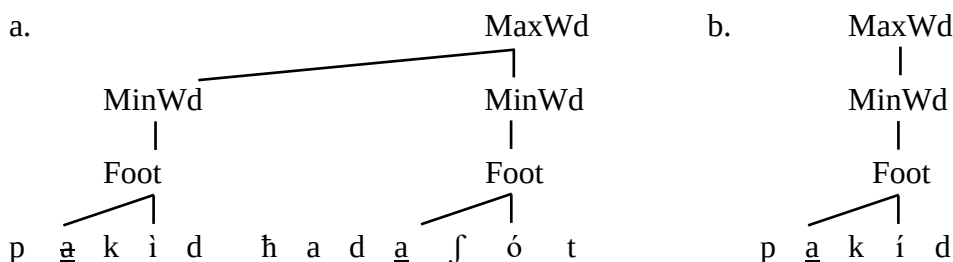
⁵⁶ Support for the single-foot analysis is found in Cohen et al. (2018), who find no secondary stress in Modern Hebrew outside Constructs.

⁵⁷ Diachronically, /a/s that resist syncope were in closed syllables, formed as a result of the gemination of the following consonant (e.g. *gan.nav*, ‘thief’). This distinction cannot be phonologically captured in MH, which does not have geminate consonants. F&T effectively propose that the historical facts has been preserved in the lexical marking of syncope vowels as ‘weak’. See Bat-El (2008) for an alternative analysis based on allomorphy.

⁵⁸ Recall that ‘face’ is pluralia tantum in Hebrew

that carries the main stress. Now, to be sure, isolated words also project to the level of the MaxWd, as shown in (97)b. Since such structures involve only one foot, however, that foot is the head foot, and the pretonic /a/ is protected from syncope for the same reason as in (97)a.

(97) Constructs vs. Isolation: /a/-syncope in non-head foot



Crucially for the present purpose, the status of a head and a dependent foot is a relative one. A foot is only a *dependent* if within its MaxWd there is another foot which heads that MaxWd. Otherwise, it is itself the head foot.

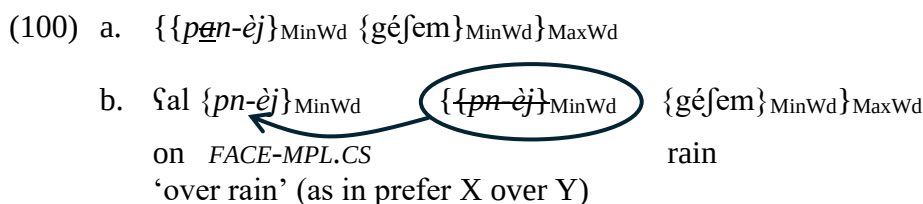
As already noted, /a/-syncope effects also hold in Pivots. Specifically, since the Pivots in (98)a-c are syntactic Construct heads, they exhibit the identical syncope effects seen in the syntactic Construct heads in (96)a-i:

(98)	<u>N (FREE)</u>	<u>PIVOT+N</u>	
a.	(ʕal) p̄an-im (on) face (pl. tantum) 'on (a) face'	ʕal= <i>pn-èj</i> géʕsem on=FACE-MPL rain '(prefer something) over rain'	cf. (96)c
b.	(be=) m̄akom (in=) place' 'in a place'	bi= <i>mkòm</i> géʕsem in=PLACE rain 'instead of rain'	cf. (96)d
c.	me=ʕ̄ajin from eye 'from an eye'	me= <i>ʕèjn</i> géʕsem from=EYE rain 'a sort of rain'	cf. (96)h

We established that syncope is associated with the syntactic head of the Construct when it is the dependent MinWd. The fact that these effects hold identically for referential syntactic Construct heads and for Pivots thus lends strong support to our claim that the Pivots, indeed, *are* the syntactic heads of the Construct, and in that syntactic position cannot form part of a constituent consisting of P+Pivot. The two questions on the table, now, are as follows:

- (99) Q1. Could incorporation *follow* syncope?
Q2. Could *syncope* follow incorporation?

For Q1 to be answered in the affirmative, prosodic structure would need to be assigned to a Construct headed by the Pivot, and that would need to be followed by movement of the Pivot to adjoin to P, under the assumption that the Pivot would be able to hold on to its prosodic status following such incorporation. This is illustrated in (100), where the Pivot would first be assigned a dependent MinWd status by being in the same MaxWd as its complement, and then move outside that MaxWd and incorporate into P:



For Q2 to be answered in the affirmative, Construct formation would need to occur *after* the Pivot adjoins to P, and it is at that point that it would be associated with prosodic structure as well as with

its bound allomorphs.⁵⁹

(101) a. $\{\{pan-PL\} \{gé\text{fem}\}_{MinWd}\}_{MaxWd}$

b. $\zeta al \{\{p\text{an-}\acute{e}j\}_{MinWd} \{p\text{an-}PL\}\}_{\{gé\text{fem}\}_{MinWd}\}_{MaxWd}$
 on FACE-MPL.CS rain
 ‘over rain’ (as in prefer X over Y)

Both these derivations, however, are problematic in at least three respects.

First, in all approaches to prosody that we know of, relative prosodic weight is evaluated where pronounced, and not where it may have originated. To exemplify, in English verb-particle constructions the particle typically carries stress if it follows the object, but that stress is lost when it is adjacent to the verb, presumably as a result of a prosodic unit having been formed between the verb and the particle: *we bróke the windows úp*; *we lét the students dówn*; vs. *we bróke up the windows*, *we lét down the student* (note also *a bréak up*, *a lét down*). Under the assumption that such expressions are derived from a common source through some form of dislocation, the different stress patterns would be mysterious, if marked prior to movement. This consideration alone militates against the derivations in (100), (101)

Second, once moved, the incorporated Pivot is in a distinct prosodic word comprising P and Pivot. As Ps do not carry stress, the Pivot, once incorporated, would automatically become the head of the prosodic word PPiv and hence would receive primary stress, and could not syncope. This, however, is contrary to fact. Just like regular Construct heads, where there are no grounds for postulating incorporation, Pivots preceding DP complements never carry primary stress and consistently syncope (where otherwise required). This cannot be accounted for under either derivation (100) or derivation (101). For derivation (100) specifically, note further that if our model were to contrive to somehow postulate the Pivot as a prosodic head as required by the incorporated structure, the result would be a prosodic head that exhibits syncope effects, an otherwise unattested situation.

Thirdly, suppose we were to entertain a solution to the two previous problems by assuming that following incorporation, under either structure, the resulting complex P, like Ps in MH in general, cannot be stressed. This is illustrated in (102). Here the Pivot acquires its prosodic status *after* incorporating into P and forming a minimal word with it. The MinWd consisting of P+Pivot now joins the following MinWd – the complement of the Pivot – to form a MaxWd, prosodically headed by the complement. (Note that under such an account it must be the Construct formed between the Pivot and its nominal complement *prior* to the incorporation that would account for the specific allomorph chosen for the MPL, *-ej*):

(102) $\{\{\zeta al \{p\text{an-}\acute{e}j\}\}_{MinWd} \{p\text{an-}ej\}\}_{\{gé\text{fem}\}_{MinWd}\}_{MaxWd}$
 on FACE-MPL.CS FACE-MPL.CS rain
 ‘over rain’ (as in ‘prefer X over Y’)

In such a derivation one would expect all PPivs to exhibit /a/ syncope regardless of the category of their complement, as the prosodic effects associated with the Pivot emerge from forming a MinWd with P, and are therefore independent of whether or not a Construct is formed with a DP complement. That, however, is not the case. When the PPiv is intransitive or has a PP complement and no Construct is formed, the Pivot occurs as an unbound form, without /a/-syncope. To illustrate, the /a/ in the PPivs (mi=)*saviv* ‘around’ or (li=)*jamin* ‘to right’ can be affected by syncope, as is illustrated in (103), when they occur with a nominal complement, and a Construct is formed. The same PPivs (mi=)*saviv* and (le=)*jamin* / (mi=)*jamin* can also occur without a complement altogether, or with a PP complement. In these contexts, no syncope is possible, as illustrated by (104)–(106).

⁵⁹ A potential variant of B would involve forming a Construct prior to incorporation, followed by incorporation which is licit only if the output structure is itself a licit Construct. The prosodic objections raised in the text hold for both variants.

- (103) a. (mi=) *səviv* ha=bájit
 (from=)AROUND DEF=house
 ‘around/in the vicinity of the house’
 b. (li=)/(mi=)*jəmin* ha=bájit /(li=)*jəmin* ha=hanhala
 (to=)/(from=)RIGHT DEF=house/(to=)RIGHT DEF=management
 ‘to the right of the house/in support of the management’
- (104) a. (mi=)*səviv* (intransitive)
 (from=)AROUND
 ‘around’
 b. (mi=)*jəmin* /(le=)*jəmin* (intransitive)
 (from=)RIGHT/ (to=)RIGHT
 ‘on the right’/‘to the right’
- (105) a. mi=*səviv* l=a=bájit
 from=AROUND to=DEF=house
 ‘around/in the vicinity of the house’
 b. (mi=)*jəmin* l=a=pina
 (from=)RIGHT to=DEF=corner
 ‘past the corner on the right’
- (106) a. *mi=*səviv* (l=a=bájit)
 b. *(mi=)*jəmin* (l=a=pina)

Before concluding, it is worthwhile noting that a weakened H-Government condition with *c-command* replacing *sisterhood* does indeed result in the availability of post-incorporation Constructs, as in the structure in (101). To the extent, then, that such Constructs are prosodically impossible, this in and of itself provides evidence against post-incorporation Construct formation, and therefore in favor of sisterhood rather than c-command as the pivotal architectural factor in H-Government as well as in Construct formation.⁶⁰

To summarize, the incorporation account must be rejected, as it gives rise to configurations that cannot be reconciled with a constrained model of prosodic representations. More specifically, the prosodic properties of the Pivot are identical to those of other (syntactic) Construct heads, in being prosodically weak, in exhibiting Construct-dependent allomorphy, and in being impacted by /*a*/ syncope. These properties are thus clearly attested without incorporation and an account which seeks to preserve them or replicate them with an incorporated Pivot is summarily unworkable.

10 Concluding comments

We began this paper by asking to what extent syntactic constituency dictates the chunks of substantive word meaning (Content). Through our study of (so-called) complex prepositions in Modern Hebrew, we hope to have shown that non-isomorphism could exist between syntactic constituents and Content units. “Words”, therefore, are neither syntactic constituents nor terminals, and the domain of Content and the domain of phonological words need not match syntactic constituency. Instead, we argued for (107), slightly modified from (2).

⁶⁰ We note as an aside that these same prosodic considerations would rule out an incorporation account that postulates mixed categories as in Panagiotidis (2010) or categorial phase extension as in den Dikken (2007). Under the assumption that within any account of incorporation P and Pivot both reside in P, then regardless of the emerging syntactic properties of the amalgam, the structure would give rise to a prosodically non-existing Construct.

Another relevant alternative to incorporation would involve M-merger as an alternative to syntactic movement (e.g. Harizanov 2014). Depending on the particular execution and its interaction with prosody on the one hand, and the domain for Content search on the other hand, such an execution may emerge as a notational variant of ours. Pending an explicit execution, we set this matter aside.

- (107) Content is not matched with a syntactic constituent or with a syntactic terminal, but with a *Span*, where a Span is a locally defined and phonologically marked string of terminals (modified from Svenonius 2012, 2016).

Spans are not constituents. Instead, they consist of two or more phonologically adjacent terminals with a well-defined relationship between them: that of H-government. At least one consequence is the existence of Content units that correlate with sequences that cross categorial boundaries. Thus, the sequence P=N in MH can form a Span and be matched with atomic Content, although N in it heads a nominal phrase from which P is excluded (see structures in (5)-(6)). By a similar logic, English combinations such as *in lieu* and *by virtue* might be Spans consisting of the boxed P=N in the schematic structure ‘ $\boxed{[P \text{ } [_{NP} N] \text{ } (of DP)]}$ ’.

For MH, the Span view carries several advantages over the competing view, according to which P+N combinations are lexically listed as such (potentially as lexical compounds) or are altogether unanalyzed P heads. First, it allows us to reduce the number of Hebrew (true) prepositions (present and past) to (at most) 9 instances, as opposed to hundreds, and dismiss the common claim, for instance in Schlezinger (1998), that P in Hebrew is an open class which enters productive word formation rules. At the very same time it allows us to fully capture the nominal properties of ‘complex prepositions’ that have been previously observed, without compromising the categorial distinction between P and N. Empirically, our structures also account for the distinct syntactic profile of simple P vs. ‘complex P’ offering a straightforward explanation to P recursion and the obligatory transitivity of the ‘simple’ vs. ‘complex’ Ps. Within our analysis, these simplifications are possible because for us, the single most significant syntactic difference between PPivs and standard PPs derives from the fact that the complement of (simple) P is not a DP, but a truncated nominal projection. Theoretically, these simplifications follow directly from the claim that word structure is syntactic, that Content is matched with some well-defined syntactic combinations, and that phonological representation is essential for en-search to proceed.

Generalizing, we hope to have shown that MH so-called complex prepositions join other morphological and morpho-phonological phenomena in reinforcing a view of the search for Content that takes as its input restricted syntactic objects, as well as phonological information. By way of a final comment, we note that the search for Content, en-search, requiring as it does phonological representation and returning specifically *substantive* meaning, complete with all vagueness, polysemy and ambiguity typically associated with conceptual knowledge, is crucially distinct from the meaning component that deals with the interpretation of constituents, structures, and functional features, which is *not* dependent on phonological representation, and which is subject to rigorous interpretation through formal semantics. If we take Content and the search for it to be part of the conceptual interface, and if we take semantic representations to belong with the intentional interface, it follows that contra Chomsky (1995 and much subsequent), there is no unified *conceptual-intentional* interface. The input to the conceptual interface passes through sound representations, the input to the intentional interface does not. The input to the intentional interface passes through a complex of formal features and functional structures; the input to the conceptual interface does not.

The replacement of syntactic constituents by syntactically-delimited Spans as the input to Content matching carries many consequences, which this paper only began to explore. We hope that the alternative perspective on Content that the proposal brings will be both useful in solving other morpho-syntactic Content mismatches and constrained enough to rule out unattested scenarios.

Appendix: more on Ps and PPivs

In section 6 we reviewed a number of structural reasons for distinguishing, syntactically, between simple P and PPivs, with the conclusion that PPivs cannot be contained in a unified P projection, whatever its internal structure. More concretely we noted that while P could be recursive PPiv cannot, and while P must be followed by a nominal complement, PPiv (as a whole) does not. In turn, we noted that these diagnostics support the conclusion that there exists, in Hebrew, a null instantiation

of P which may precede Pivots, giving rise to ‘bare’ PPivs.

In this appendix we return to some aspects of that discussion. In section A1, we discuss another important difference between ‘simple’ P and PPivs involving the distribution of complementizers. In section A2 we address a potential problem involving the occurrence of the complementizer *je=* following bound Construct heads. Finally, in section A3 we offer a few comments on two cases in which the distribution of P and PPiv is the same (and distinct from that of Ns), in apparent contrast with other diagnostics thus far offered.

A1 Complementizer Selection

A1.1 Complementizer *je=*

The complementizer *je=* occurs in MH preceding tensed sentential clauses, including within headed relative clauses. As illustrated in (A1), it can occur following PPivs:

- (A1) a. *li=fn-ej* *je=ha=mora* *niχnes-a* *l=a=kita*
 to-FACE-MPL.CS COMP=DEF=teacher enter-3FSG to=DEF=classroom
 ‘before the teacher entered the classroom’
 b. *bi=glal* *je=ha=mora* *niχnes-a* *l=a=kita*
 in-TURD COMP=DEF=teacher enter-3FSG to=DEF=classroom
 ‘because the teacher entered the classroom’
 c. *ʃal=men-at* *je=ha=mora* *t-ikanes* *l=a=kita*
 on=PORTION-FSG.CS COMP=DEF=teacher 3FSG-enter.FUT to=DEF=classroom
 ‘in order for the teacher to enter the room’

However, the complementizer *je=* cannot occur after a simple P (A2). Once an expletive *ze* ‘it’ is inserted as a complement of P, however, the result is licit:⁶¹

- (A2) a. *le=*(ze) je* *ha=mora* *niχnes-a/t-ikanes* *l=a=kita*
 to=*(it) COMP DEF=teacher enter-3FSG/3FSG-enter.FUT to=DEF=classroom
 ‘to the fact that the teacher entered/will enter the classroom’
 b. *mi=*(ze) je*
 from=*(it) COMP
 c. *be=*(ze) je*
 in=*(it) COMP
 d. *ʃal *(ze) je*
 on COMP
 e. *ʔel *(ze) je*
 toward *(it) COMP
 f. *ʃim *(ze) je*
 with *(it) COMP

A1.2 Complementizer *ʔafer* (higher register)

The complementizer *ʔafer* is excluded preceding tensed sentential complements in MH but is licit as a higher register complementizer for (headed) tensed relative clauses as in (A3), where it alternates with *je=* (A3).

⁶¹ As noted (fn 6), we follow Gesenius (1910) (among others) in taking the *k-* occurring e.g. in *k/je=* (‘when’), *ka=ʔafer* (‘when’) to be homophonous with a complementizer. The preposition *mi(n)* ‘from’ can occur preceding *je=* sentential comparatives (i). We assume without further discussion that the structure of comparatives may involve distinct restrictions from those associated with regular P-CP combinations, thereby allowing *mi=je=* in comparatives, but not, for example, in (A2).

i. *hi* *joter* *gvoha* *mi=je=χafav-ti*
 she more tall from=COMP=thought-1SG
 ‘she is taller than I thought’

- (A3) a. ha=susim ?afer/fe=nimtse?-u haju fajax-im le=flómo
 DEF=horses COMP=find.PASS-3MPL were belong-PL to=Shlomo
 ‘the horses that were found belonged to Shlomo’
 b. ha=susim ?afer/fe=matsá-t-em haju fajax-im le=flómo
 DEF=horses COMP=find-2-MPL were belong-PL to=Shlomo
 ‘the horses you stole belonged to Shlomo’

When we turn to prepositions, we find that *?afer* can occur directly after simple prepositions, as in (A4), but is excluded following PPiv (A4). Note that in these cases what is embedded under the preposition is a *headless* relative clause:⁶²

- (A4) le=?afer me=?afer be/ba=?afer ?al=?afer ?el=?afer ?ad=?afer⁶³
 to=COMP from=COMP in=COMP on=COMP toward=COMP until=COMP
 to from where about toward until
 (A5) *{li=fn-ej mi=p?-at bi=glal ?al=?af} ?afer
 {to=FACE-MPL.CS from-EDGE-FCS in-TURD to=NEG} COMP
 {before because because despite} that

A1.3 Complementizers distribution : a grammatical fragment

Recall that we postulated an obligatory transitivity for P (see (49) in section 6.4 and related discussion). The ungrammaticality of *fe=* after ‘simple’ prepositions follows then from the fact that *fe=* introduces tensed clauses and not nominal projections. Not so for PPivs, where the Pivot is already a projection of N thereby satisfying (49) and making the occurrence of a subsequent clause immaterial from this perspective. As such, the distribution of *fe=* provides additional evidence that the Pivot is syntactically N, and not (part of) P.

Turning to the distribution of *?afer*, recall that it is restricted to relative clauses, by assumption a nominal projection. The availability of such relative clauses after simple prepositions as in (A4) is therefore predicted. The exclusion of *?afer* following Pivots, however, is a trickier matter, and a full explanation would require delving into the minutia of MH relative clause structure, which cannot be undertaken here for reasons of space. In a nutshell, we note that the *?afer* clauses in (A4)-(A5) are *headless* relatives, and it is exactly such headless relatives that are excluded following PPiv. *Headed* relatives, with either *fe=* and *?afer*, are licit both for Ps and PPivs:

- (A6) {li=fn-ej bi=glal la=mrot} ha=séfer fe=/?afer kaní-ti
 {to=FACE-MPL.CS in=★ to=★ DEF=book COMP bought-1SG
 {before because despite the book that I bought...’
 (A7) {l b k ?al mi ?el ?im} ha=séfer fe=/?afer kaní-ti
 {to in like on from toward with} DEF=book COMP bought-1SG
 {to in like on from toward with} the book that I bought...’

The contrast between the ungrammatical (A5) and the grammatical (A6) therefore seems to stem from the absence of an overt nominal head in the former, vs. the latter. We would therefore like to suggest that the impossibility of embedding headless *?afer* relatives under PPivs is a direct result of the absence of any overt N which the Pivot can construct with. Concretely, we postulate the restriction in (A9), reducible to the fact that the complement of the Construct is the prosodic head. As the structure in (A9) lacks a prosodic head a Construct cannot be formed.

- (A8) [D (Ø) [CP ?afer ... [....]]]

⁶² A handful of cases of *?afer*+Pivot occur in Biblical Hebrew (PPiv *?afer*/PPiv total): *táhat ?afer* ‘under’ (13/247); *?ahr-ej ?afer* ‘after’ (8/292) *mi-l-vad ?afer* ‘except’ (4/29); *le-má?an ?afer* ‘for’ (11/235). But *li-fn-ej ?afer* ‘before’ (0/636); *le-jad ?afer* ‘next to’ (0/13). These are all strictly ungrammatical in MH. Compare, as well, with BH *la-?afer* (23), *ba-?afer* (17), *me-?afer* (19), *?al ?afer* (29), *?ad ?afer* (47).

⁶³ *?im ?afer*, ‘with that’, is marginal, and possibility altogether illicit for some. It also occurs in BH only 4 times (as compared to 20-40 for other P-*?afer* combinations). Recall as well that *?im* does not occur within PPivs.

(A9) *[_{CS} N [_{DP} Ø]]

This account, sketchy though it might be, crucially relies on the formation of a Construct between the Pivot, itself a nominal projection, and its complement. If the Pivot is a part of a prepositional head, and if (as in Botwinik-Rotem 2008a,b) all prepositions construct with their complements, it would certainly be plausible to propose that the same prosodic constraint could apply to the Construct formed between the preposition and its complement. However, such an execution would rule out not only PPiv+headless relatives but P+headless relatives, clearly the wrong result.

It is worthwhile noting that while the majority of ‘bare’ PPiv tokens (17/25 in our corpus) favour a DP complements exclusively. The three that allow a complementizer (including *bejn*, ‘between, among’ see fn. 33) are in (A10), where they comfortably occur with *je=*, but not with *ʔafer*, on a par with PPivs and differently from Ps, lending additional support to their status as such:⁶⁴

- (A10) a. *ʔahr-ej* *je=* /**ʔafer*
 AFTER-MPL.CS COMP
 ‘after’
- b. *bejn* *je=* /**ʔafer* *ʔamar-t* *le=bejn* *je=* /**ʔafer* *lo* *ʔamar-t*
 BETWEEN COMP said-2F to=BETWEEN COMP NEG said-2F
 ‘whether you said it or not’
- c. (be=)*vadaj* *je=* /**ʔafer*
 (in=)CERTAINTY COMP
 ‘it is certain that’

A2 PPiv + *je=*

As the reader has no doubt noticed, in the licit cases in (A1) and equivalents where PPiv is followed by a sentential complement with *je=*, the Pivot is followed by a complementizer and not by an instance of N, thereby requiring an account for the emergence of a bound CS form. Note, in particular, the bound form of the MPL Construct head in (A11)a,b, the bound form of the feminine suffix in (A11)d, and the syncope in (A11)c,d.

- (A11) a. *li=fn-ej* *je=*
 to=FACE-MPL.CS COMP
 ‘before’
- b. *ʔaxr-ej* *je=*
 BACKSIDE-MPL.CS COMP
 ‘after’
- c. *bi=mkom* *je=*
 in=PLACE COMP
 ‘instead’
- d. *ʕal=mn-at* *je=*
 on=PORTION-FSG.CS COMP
 ‘in order’

A closer examination of the complementizer *je=* in PPiv, however, reveals that it has distinct prosodic properties from those attested when it follows a non-Pivot N, V or A. In these latter cases, *je=* is a proclitic, which is to say that it joins the prosodic word (PrWd) to its right. However, when

⁶⁴ *ʕad* presents an unusual picture. We noted already in fn 33 that it can be both P and Pivot. In an added peculiarity, it is the only case we are aware of in which a P-like element can be followed by both *je=* and *ʔafer* and where *ʔafer* actually introduces a complement clause, and not a RC. We set this matter aside.

- i. a. *tsaʕád-nu* *ʕad* *je=ha=femeʃ* *zarha*
 march-1PL till COMP=the-sun came.up
- b. *tsaʕád-nu* *ʕad* *ʔafer* *ha=femeʃ* *zarha*
 march-1PL till COMP DEF=sun came.up
 ‘we marched until sunrise’

following a Pivot, its properties tally with those of an *enclitic* joining the PrWd to its left, to the Pivot, that is, thereby giving rise to a discernibly distinct (optional) intonational contours.

- (A12) a. ha=maskana je=ha=knija hajta mejutéret
 DEF=conclusion COMP=DEF=purchase was unnecessary
 ‘the conclusion that the purchase was unnecessary’
 b. barur je=ha=knija hajta mejutéret
 obvious COMP=DEF=purchase was unnecessary
 ‘it’s obvious that the purchase was unnecessary’
 c. hi t-avin je=ha=knija hajta mejutéret
 she 3FSG-understand.FUT COMP=DEF=purchase was unnecessary
 ‘she will understand that the purchase was unnecessary’
- (A13) a. mi-pn-éj =je ha=knija hajta mejutéret
 from-FACE-MPL.CS =COMP DEF=purchase was unnecessary
 ‘because DEF=purchase was unnecessary’
 b. řal=mn-át =je ha=knija titbatséař
 on=PORTION-F.CS =COMP DEF=purchase 3FSG-occur.FUT
 ‘in order for the purchase to occur’
 c. bi=mkóm =je ha=knija t-itbatséař
 in=PLACE =COMP DEF=purchase 3FSG-occur.FUT
 ‘instead of the purchase occurring’

The assumption that the contrast in (A12)-(A13) is between an enclitic vs. proclitic finds support in the coordination facts in (A14)-(A15), where the coordination of two PPiv+/je is licit, but not so the coordination of {V,A,N}+/je, and where the complementizer must follow the coordination.⁶⁵

- (A14) a. řal mn-át=je ve bi=glál=je
 on PORTION-F.CS=COMP and in=TURD=COMP
 ‘in order to and because of’
 b. li=fn-éj=je ve ke=děj=je ve řaf řal pí=je
 to=FACE-MPL.CS=COMP and as=ř.CS=COMP and too on MOUTH=COMP
 ‘before (that) and in order that, and despite (that)’
- (A15) a. řařáv-ti (*je) ve řamár-ti (je) *V-je & V-je/✓V&V-je
 think-1SG (*COMP) and say-1SG (COMP)
 ‘I thought that and I said that...’
 b. ha=pirsum (*je) ve ha=maskana (je) *N-je & N-je/✓N&N-je
 DEF=publication (*COMP) and DEF=conclusion (COMP)
 ‘the publication that and the conclusion that’
 c. *barur (*je) ve ratsuj (je) *A-je & A-je/✓A&A-je
 clear (*COMP) and desirable (COMP)
 ‘it is clear (*that) and desirable (that)’

The converse holds as well. While two Vs, N’s and A’s could coordinate and share a common je complement, such coordination is barred for Pivots (*Pivot&Pivot je):

- (A16) a. *ke=děj ve bi=l=řad-ej je
 as=ř.CS and in=to=ETERITY-PL.CS COMP
 ‘in order (to) and without that’
 b. *mi=řum ve řal=mn-at je
 from=ONION and on=PORTION-FSG.CS COMP
 ‘because and in order that’

We therefore conclude that the bound effects associated with Pivot when followed by je are induced

⁶⁵ The availability of (A14) and similar was pointed to us by Ur Shlonsky (p.c.).

by that proclitic. As such, *fe*'s behavior is on a par with other proclitics that may join with the Pivot, some of which are illustrated in (A17)-(A19):⁶⁶

- (A17) *li=fn-éj=χen* *ʔahr-éj=χen*
to-FACE-MPL.CS=so BEHIND-MPL.CS=so
'before' 'after'
- (A18) *ʔahár=kaχ* *béjn-ko* *mi=l=vád=kaχ* *be=ʔél=kaχ* (also *be=ʔel-káχ*)
BEHIND=so BETWEEN=so from=to= ≠=so in=OF=so
'later' 'anyway' 'except that' 'because of that'
- (A19) *li=fn-éj=ze* *bi=glál=ze* *bi=l=ʔad-éj=ze* *bi=mkóm=ze...*
to-FACE-MPL.CS=this in=TURD=this in=to=ETERNITY in=PLACE=this
'before that' 'because of that' 'without it' 'instead of this'

A3. Caveats

In two domains the behavior of PPiv tallies with that of prepositions and not with that of nominals. First, like simple Ps and unlike Constructs heads, Pivots freely construct with proper names. N-PN Constructs are not strictly ungrammatical but are strongly dispreferred in MH in possessive contexts. The relevant cases are in (A20)a–c. Secondly, pronominal suffixes on (referential) Ns have moved to a higher register in MH and have virtually disappeared from the spoken language. Like simple Ps, and unlike referential Ns, Pivots freely take pronominal complements without register shift, cf. in (A21)a–c.

- (A20) a. P [Pr.Name] b. P=Pivot [Pr.Name]
tarám-ti *le=ʔódeh* *tarám-ti* *le=máʔan* *ʔódeh*
donate-1SG to Odeh donate-1SG to=ADDRESS Odeh
'I donated to Odeh' 'I donated to support Odeh'
- c. P-DP-[Pr.Name]
ʔaláx-ti *ʔet* *ha=taʃlum* *^Hle=maʃan* *ʔódeh^H*
send-1SG acc DEF=payment to=address Odeh
'I sent the payment to Odeh's address'
- d. P-DP *ʔel* [Pr.Name]
ʔaláx-ti *ʔet* *ha=taʃlum* *l-a=maʃan* *ʔel* *ʔódeh*
send-1SG acc DEF=payment to=DEF=address OF Odeh
'I sent the payment to Odeh's address'
- (A21) a. P [Pron] b. P=Pivot [Pron]
tarám-ti *l-o* *tarám-ti* *le=maʃan-o*
donate-1SG to-3MSG donate-1SG to=address-3MSG
'I donated to him' 'I donated to his cause'
- c. P-DP-[Pron]
ʔaláx-ti *ʔet* *ha=taʃlum* *^Hle=maʃan-o^H* (*ha=prati*)
sent-1SG ACC DEF=payment to=address-him (DEF=private)
'I sent the donation to his private address'

⁶⁶ A prosodic puzzle is nonetheless raised by these Pivot-pronominal forms. As explained in section 9.2, nominal suffixes typically attract stress, thereby triggering the /a/-syncope in the stem (i) which is also found in the Construct, as illustrated by the bare Pivot in (ii). Enclitics, as noted, do not attract stress, but the syncope in the Pivot still applies (iii):

i. <i>savív</i>	→	<i>saviv-á</i>	(ii)	<i>savív</i>	<i>ha=konflíkt</i>	(iii)	<i>svív=ze</i>
around		around-f		around	def=conflict		around=this
'around (adv.)'		'vicinity'		'about the conflict'			'about this'

The matter, we note, is not unique to PPivs, but marks a host of pronominal clitics, thereby suggesting that some adjustment to the prosodic model is needed, a task we will set aside for future research.

d. P-DP *fel* [Pron]

ʃaláx-ti ʔet ha=taʃlum l-a=máʃan (ha=prati) *fel-o*
 sent-1SG ACC DEF=payment to=DEF=address (DEF=private) OF=3MSG
 ‘I sent the donation to his private address’

We believe the effects attested in (A20)-(A21) are due to [D[DP]] Constructs (R-Constructs) with a *possessum-possessor* relation having fallen out of favor in MH, to be replaced with an analytic form using the grammatical marker *fel* (roughly ‘of’). At the same time, neither [D[NP]] Constructs (M-Constructs) nor Pivot-DP Constructs can support the marker *fel*, which requires referentiality for both *possessum* and *possessor* thereby forcing both to be DPs. Proper name complementation and pronominal suffixes have as a result survived in both [D[NP]] M-Constructs and in [N[DP]] Pivot Constructs as they are the only contemporary felicitous option for marking them in genitive contexts.⁶⁷

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⁶⁷ As observed at least as early as Rosen (1966), residual uses of [D[DP]] Constructs in MH obey distributional generalizations which are not easy to characterize. For instance, they remain fully grammatical when they head a deverbal nominal (*fomer ha=masaḥ* ‘guard of the screen’ *haḥṣat ha=ʕir* ‘bombing of the city’), and they favor a plural head over a singular one (*sifr-ej ha=jéled* ‘the child’s books’, *ʕijur-ej van goḥ* ‘Van Gogh’s paintings’ vs. *ʕséfer ha=jéled* ‘the child’s book’, *ʕijur van-goḥ* ‘Van Gogh’s painting’). We have little to offer here on these facts beyond observing their existence, but see Danon (2018) for further discussion.

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